

TESTS OF MISSING COMPLETELY AT RANDOM BASED ON SAMPLE COVARIANCE MATRICES

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We study the problem of testing whether the missing values of a potentially high-dimensional dataset are Missing Completely at Random (MCAR). We relax the problem of testing MCAR to the problem of testing the compatibility of a collection of covariance matrices, motivated by the fact that this procedure is feasible when the dimension grows with the sample size. Our first contributions are to define a natural measure of the incompatibility of a collection of correlation matrices, which can be characterised as the optimal value of a Semidefinite Programming (SDP) problem, and to establish a key duality result allowing its practical computation and interpretation. By analysing the concentration properties of the natural plug-in estimator for this measure, we propose a novel hypothesis test, which is calibrated via a bootstrap procedure and demonstrates power against any distribution with incompatible covariance matrices. By considering key examples of missingness structures, we demonstrate that our procedures are minimax rate optimal in certain cases. We further validate our methodology with numerical simulations that provide evidence of validity and power, even when data are heavy tailed. Furthermore, tests of compatibility can be used to test the feasibility of positive semidefinite matrix completion problems with noisy observations, and thus our results may be of independent interest.

1. Introduction. Incomplete data are a common occurrence in almost all areas of statistical application, and the mechanisms leading to such data are diverse. For example, subjects in a survey may choose not to respond to certain questions, leading to missing values, or a practitioner may wish to combine data collected in different studies, where different variables were recorded in each. With incomplete data, traditional approaches become unreliable or even inapplicable, leading to a significant effect on the conclusions that can be drawn from the data. The most common approaches to dealing with missing values are to remove any incomplete observations, and thus to perform a complete-case analysis, or to replace any missing entry with a representative value, using an imputation method (e.g., [Yates \(1933\)](#), [van Buuren and Groothuis-Oudshoorn \(2011\)](#), [Stekhoven and Bühlmann \(2011\)](#)). However, the validity of such procedures, and the choice of an appropriate one, depends crucially on the mechanism that determines the missingness. Mechanisms have traditionally been classified as Missing Completely At Random (MCAR), Missing At Random (MAR) and Missing Not At Random (MNAR) (e.g., [Little and Rubin \(2002\)](#)) according to the dependence structure between the variables themselves and their missingness, with such assumptions being required to link observations to targets of inference.

The typical formal setting is to suppose that we observe independent and identically distributed copies of a random object $X \circ \Omega$, where X takes values in some product space

Received December 2024; revised May 2025.

MSC2020 subject classifications. 62G10.

Key words and phrases. Missing data, missing completely at random, minimax testing, semidefinite programming.

$\mathcal{X} = \prod_{j=1}^d \mathcal{X}_j$, where Ω takes values in $\{0, 1\}^d$ and where we define the operator $\circ$ by

$$(x \circ \omega)_j = \begin{cases} x_j & \text{if } \omega_j = 1, \\ \text{NA} & \text{if } \omega_j = 0. \end{cases}$$

The assumptions named above then control the dependence between the data X and the missingness indicator Ω . The simplest case of MCAR is when these are independent, denoted $X \perp\!\!\!\perp \Omega$, so that the data we observe is representative of the population, even if it is incomplete.¹ Under MCAR, we can often employ statistical methodologies that are easy to interpret and make good use of all incomplete data, with solid theoretical guarantees having been developed in various modern statistical problems such as high-dimensional regression (Loh and Wainwright (2012)), high-dimensional or sparse principal component analysis (Zhu, Wang and Samworth (2022), Elsener and van de Geer (2019)), classification (Cai and Zhang (2019), Sell, Berrett and Cannings (2024)), and precision matrix and changepoint estimation (Follain, Wang and Samworth (2022)). MCAR also allows for the use of a simple complete-case analysis which, in certain cases, such as when we have small sample sizes, can be preferable to complex procedures (e.g., Aleksić, Cuparić and Milošević (2023)). However, if MCAR does not hold, which is common in practice, alternative methods may be required.

Hypothesis tests can be used to guide practitioners in deciding whether or not missingness assumptions are reasonable. The goal of this work is to study the problem of testing the hypothesis of MCAR, which has been the subject of much research in the missing data literature. Most prior work has been developed within the context of parametric models. For example, Little (1988) works under the hypothesis that the data are Gaussian in the setting that all pairs of variable are observed together (see Section 6 for further details). Fuchs (1982) considers discrete data in the setting that a large number of complete cases are available. In both cases, the methods are likelihood ratio tests, with the MLEs calculated using the EM algorithm (Dempster, Laird and Rubin (1977)) and validity and power guarantees based on classical asymptotics. More recently, Berrett and Samworth (2023) provided a nonparametric formulation of the problem and methodology that was proved to be widely powerful under minimal assumptions. The key insight of Berrett and Samworth (2023) is to relate the problem of testing MCAR to the problem of testing *compatibility*, for which we now recall the definition. For $S \subseteq [d] := \{1, \dots, d\}$, denote by $\{\Omega = \mathbb{1}_S\}$ the event that X_j is observed if and only if $j \in S$, write $\mathbb{S} = \{S : \mathbb{P}(\Omega = \mathbb{1}_S) > 0\}$ for the set of all possible observation patterns and write P_S for the distribution of the observation $X_S | \{\Omega = \mathbb{1}_S\}$. We say that the collection $(P_S : S \in \mathbb{S})$ is compatible if there exists a distribution P on $\mathcal{X}$ with marginal distribution P_S on $\mathcal{X}_S$ for all $S \in \mathbb{S}$. Under MCAR, the distribution P_S is equal to the marginal distribution of the population distribution $\mathcal{L}(X)$ on $\mathcal{X}_S := \prod_{j \in S} \mathcal{X}_j$, so it must be the case that $(P_S : S \in \mathbb{S})$ is compatible. Hence, if $P_{\mathbb{S}} := (P_S : S \in \mathbb{S})$ is *incompatible*, then the data cannot be MCAR. In fact, it is shown (Berrett and Samworth (2023), Proposition 1) that this reasoning is tight in that it is not possible to rule out MCAR based on observations of $X \circ \Omega$ if $P_{\mathbb{S}}$ is compatible. In general, fully testing the compatibility of a collection of distributions requires us to look at complex interactions between the distributions, and methods for doing so will have sample complexity that is exponential in the dimension d . Our work aims to provide methods that are valid and powerful without strong assumptions while being effective as the dimension grows.

Our methodology will be based on testing the compatibility of collections of covariance matrices, which can be estimated consistently even for large d . Earlier studies have employed the covariance matrix to assess MCAR. As briefly discussed above, Little (1988) studied a

¹This scenario is sometimes referred to as *everywhere MCAR*, which should not be confused with *realised MCAR* (see Seaman et al. (2013) for a more detailed discussion on this distinction).

likelihood ratio test of MCAR, effectively examining the homogeneity of means and covariances under the assumption of normality. However, Little expressed scepticism about its effectiveness unless the sample size is exceptionally large and the assumption of normality holds. This scepticism was further validated in simulations by [Kim and Bentler \(2002\)](#), who also developed a test for consistency of means and covariances based on generalised least squares. Both of these approaches work by comparing the sample covariance matrix associated to a given missingness pattern to the corresponding submatrix of an estimated complete covariance matrix. More recently, [Jamshidian and Jalal \(2010\)](#) developed k -sample tests of the equality of covariance matrices, given complete data, based on Hawkins' test ([Hawkins \(1981\)](#)). Using empirical evidence, they then argued that these tests could be combined with imputation techniques to test the homogeneity of covariance matrices calculated using incomplete data. These methodologies can be effective when the corresponding assumptions are met and when a complete covariance matrix can be consistently estimated.

Our method works by directly checking the compatibility of the observed sample covariance matrices, making no assumptions on the form of the underlying distributions and not requiring the estimation of a complete covariance matrix. In particular, this second point means that our test can be applied with any collection $\mathbb{S}$ of missingness patterns. More precisely, at the population level, we will consider $\Sigma_{\mathbb{S}} = (\Sigma_S : S \in \mathbb{S})$, a collection of suitably-normalised covariance matrices Σ_S associated to the law of $X_S|\{\Omega = \mathbb{1}_S\}$, and design a statistical test to check if $\Sigma_{\mathbb{S}}$ is compatible, meaning that each Σ_S can be obtained by marginalising a general $d \times d$ positive-semidefinite matrix Σ , that is, $(\Sigma)_S = \Sigma_S$. If MCAR holds then for each $S \in \mathbb{S}$, we must have $\text{Cov}(X_S|\Omega = \mathbb{1}_S) = (\text{Cov}(X))_S$, the block of the covariance matrix of X corresponding to the variables in S , so that the collection $(\Sigma_S : S \in \mathbb{S})$ must be compatible. Hence, if we can reject the hypothesis $H_0 : \Sigma_{\mathbb{S}} \text{ compatible}$, then we can reject the hypothesis of MCAR. See [Figure 1](#) for a pictorial summary of the key concepts so far.

More generally, one can consider the problem of testing the compatibility of moments of order $p \geq 1$ and, if it is found that these moments are incompatible, one can reject MCAR. For $p = 1$, this problem reduces to testing the compatibility of mean vectors, which essentially boils down to testing the equality of means. This has been studied in the statistical literature for over a century, and we refer to existing methods for solving this problem (e.g., [Wilks \(1946\)](#), [Little \(1988\)](#)). In order to have power against a wider range of alternatives, while limiting the complexity of the testing procedure, we restrict attention in this work to the natural $p = 2$ problem. Here, there are still various ways in which compatibility can fail. For example, we can rule out H_0 if $\Sigma_{\mathbb{S}}$ is *inconsistent*, in the sense that there are two observation patterns $S_1, S_2 \in \mathbb{S}$ for which $(\Sigma_{S_1})_{S_1 \cap S_2} \neq (\Sigma_{S_2})_{S_1 \cap S_2}$, meaning that there exists a pair of variables whose covariance takes different values in different observation patterns. Testing the consistency of covariance matrices reduces to testing the equality of smaller covariance matrices, which has again been previously studied (e.g., [Hawkins \(1981\)](#)). The corresponding nonparametric problem of testing the consistency of distributions was studied by [Li and Yu \(2015\)](#), [Spohn et al. \(2025\)](#) in the context of testing MCAR. However, it is possible to test more than consistency. As a concrete example, consider the case where $d = 3$, where $\mathbb{S} = \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$, and where

$$\Sigma_{ij} = \begin{pmatrix} 1 & \rho_{ij} \\ \rho_{ij} & 1 \end{pmatrix}$$

with $\rho_{23} = \rho_{13} = -\rho_{12} = \rho$. Then $\Sigma_{\mathbb{S}}$ is compatible if and only if $\rho \leq 1/2$, even though it is always consistent. This is because the only matrix Σ such that $(\Sigma)_S = \Sigma_S$ for all $S \in \mathbb{S}$ is

$$\Sigma = \begin{pmatrix} 1 & -\rho & \rho \\ -\rho & 1 & \rho \\ \rho & \rho & 1 \end{pmatrix},$$

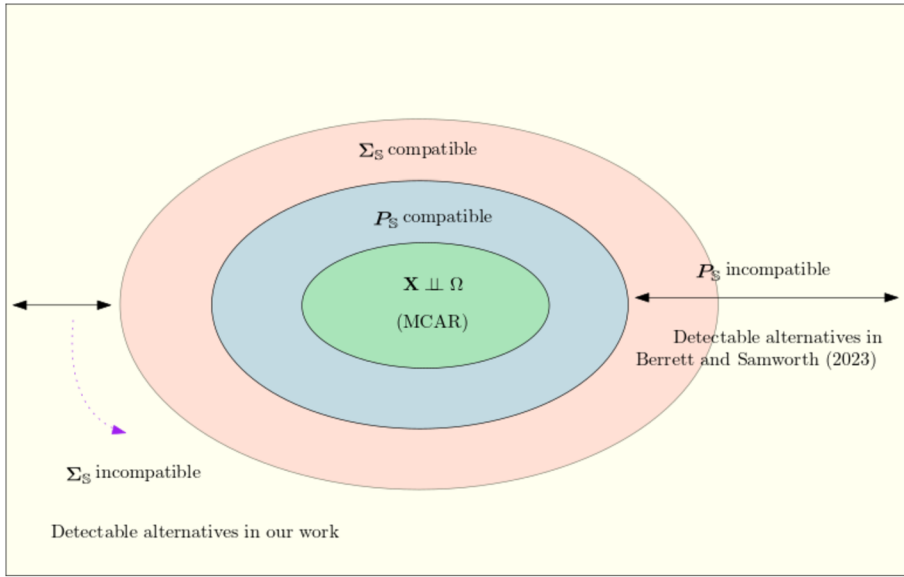


FIG. 1. Our framework. We relax the methodology in Berrett and Samworth (2023) and consider (suitably normalised) covariance matrices instead of full distributions. The price we pay is to create an extra ring (red area) in which we cannot detect departure from H_0 just by looking at Σ_S . For example, if the collection of third-moment tensors were inconsistent, but Σ_S were compatible, we would not be able to reject MCAR, although P_S would be incompatible.

whose eigenvalues are $1 + \rho$ and $1 - 2\rho$, which therefore is not positive semidefinite if $\rho > 1/2$.

The above example is relatively simple because any pair of variables is observed together so that the full covariance matrix can be estimated. However, we can characterise compatibility for any S (see Proposition 4). While the compatibility of distributions can be characterised using linear programming (e.g., Kellerer (1984)), characterising the compatibility of covariance matrices requires ideas from semidefinite programming (SDP), which studies linear optimisation problems over spectrahedra (e.g., Blekherman, Parrilo and Thomas (2012), Vandenberghe and Boyd (1996)). If Σ_S is consistent, compatibility is equivalent to the feasibility of a positive semidefinite matrix completion problem, where we observe a partial symmetric matrix $A = (a_{ij})$ for positions (i, j) in a certain set of edges, and aim to construct a positive semidefinite completion of A . This problem is extensively studied owing to its widespread applications in diverse fields such as probability, statistics, systems engineering and geophysics; see, for example, Laurent (2009) and the references therein for an introduction to the topic. Statistical questions associated with such problems are relatively underexplored, though we mention the recent work Waghmare and Panaretos (2022) that provides estimated completions of covariance operators in settings where completions always exist. A distinct but related problem that has received more attention in the statistics literature is low-rank matrix completion. In particular, significant contributions (Candès and Recht (2009), Candès and Tao (2010), Recht (2011)) have been made in the realm of convex optimization, where a low-rank matrix is recovered from partial observations after introducing a nuclear norm penalty. In our work, we make no low-rank assumptions and our main interest is in answering the question of whether or not a positive semidefinite completion exists.

We now briefly outline our main contributions. In Section 2, we introduce a data-driven bootstrap test for MCAR, detailed in Algorithm 1. We also state our main result (Theorem 1), that this procedure is uniformly valid over expanding subsets of the null hypothesis, excluding distributions close to the boundary, and uniformly powerful against alternatives separated

from the null hypothesis. Additionally, we establish its asymptotic validity over the entire null hypothesis in a specific example (Proposition 2). This test is based on a numerical measure, $R(\Sigma_{\mathbb{S}})$, which quantifies the incompatibility of a collection of correlation matrices $\Sigma_{\mathbb{S}}$. We define this measure in Section 3 and establish key properties, including a useful and interpretable dual representation (Proposition 5). In Section 4, we shift our focus to the empirical estimation of this index and analyse its concentration properties. Assuming nonsingularity of the correlation matrices, $\Sigma_S \succeq cI_{|S|}$ for all $S \in \mathbb{S}$, we present an oracle test based on knowledge of $c > 0$, and we state a result on its validity and power (Theorem 7). This result is crucial for the proof of Theorem 1. In Section 5, we analyse the performance of our oracle test in various examples and show that its separation rate is near-minimax optimal in some cases, while studying properties of the associated semidefinite programmes. In Section 6, we validate our methodology in numerical experiments. Proofs of all of our results, as well as auxiliary results, are deferred to the Supplementary Material A (Bordino and Berrett (2025)).

We conclude the Introduction with some notation that is used throughout the paper. In general, we will denote covariance matrices by Ω and correlation matrices (or other suitably normalised covariance matrices) by Σ . For $d \in \mathbb{N}$, we write $[d] := \{1, \dots, d\}$, and indicate with $|B|$ the cardinality of the set B . Given $a, b \geq 0$, we write $a \lesssim b$ to mean that there exists a universal constant $C > 0$ such that $a \leq Cb$. We use $a \wedge b$ for $\min\{a, b\}$, and $a \vee b$ for $\max\{a, b\}$. We will denote with $\mathbf{0}_d$ the null vector of dimension d , with $\mathbf{1}_d$ the all-one vector, with $\mathbf{e}_j$ the j th element of the canonical basis of $\mathbb{R}^d$, with $\mathbf{O}_{d_1, d_2}$ the zero matrix of dimension $d_1 \times d_2$, with $\mathbf{O}_d := \mathbf{O}_{d, d}$, and with I_d the identity matrix of dimension d . We will omit the subscript with the dimension d when it is clear from the context. For symmetric matrices A, B of dimension d , we write $A \succeq 0$ to mean that A is positive semidefinite, write $A \succeq B$ to mean that $A - B \succeq 0$, write $\text{diag}(A)$ to indicate the vector whose elements coincide with the diagonal entries of the matrix A , and $\text{diag}(\mathbf{v})$ for a vector $\mathbf{v} = (v_1, \dots, v_d)$ to indicate a diagonal matrix with diagonal elements equal to v_i . We will indicate the trace of A with $\text{tr}(A)$, the determinant with either $|A|$ or $\det(A)$, and the minimum and maximum eigenvalues of A with $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$, respectively. We use $\|\cdot\|_p$ for the l_p -norm of a vector. We will use $\|\cdot\|_*$ for nuclear norm, or Schatten-1 norm of a matrix, $\|\cdot\|_2$ for the spectral norm, and $\|\cdot\|_F$ for the Frobenius norm. For random elements X, Y , we write $X \perp\!\!\!\perp Y$ to mean that X and Y are independent. For $\sigma > 0$, a random variable X with mean $\mu = \mathbb{E}[X]$ is said to be σ -sub-Gaussian if

$$\mathbb{E}[e^{\lambda(X-\mu)}] \leq e^{\sigma^2 \lambda^2 / 2} \quad \text{for all } \lambda \in \mathbb{R},$$

while, for $(\nu, \alpha) \in (0, \infty)^2$, it is said to be (ν, α) -subexponential if

$$\mathbb{E}[e^{\lambda(X-\mu)}] \leq e^{\frac{\nu^2 \lambda^2}{2}} \quad \text{for all } |\lambda| < \frac{1}{\alpha}.$$

A random vector X in $\mathbb{R}^n$ is said to be σ -sub-Gaussian if every one-dimensional projection, that is, $\mathbf{v}^T X$ with $\mathbf{v} \in \mathbb{R}^n$ and $\|\mathbf{v}\| = 1$, is σ -sub-Gaussian in the sense defined above.

2. Statistical setting and bootstrap test of MCAR. We aim to test $X \perp\!\!\!\perp \Omega$ by examining the collection of suitably normalised covariance matrices across different missingness patterns, and we will see in the sequel (e.g., Section 5) that the structure of these patterns has a strong influence of the complexity of the problem. Instead of assuming that we have access to i.i.d. copies from $X \circ \Omega$, it is convenient to condition on the realisations of Ω , so that the collection of patterns $\mathbb{S}$ and the samples sizes $(n_S : S \in \mathbb{S})$ are fixed. This does not impose a significant constraint, and all the theoretical results can be adapted to the unconditional model. We then assume that for each $S \in \mathbb{S}$ we have access to an independent sample

$$X_{S,1}, \dots, X_{S,n_S} \stackrel{\text{i.i.d.}}{\sim} P_S$$

Algorithm 1 MCAR bootstrap test checking compatibility of correlation matrices

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1: Given data  $X_{\mathbb{S}}$ , discard all patterns  $S \in \mathbb{S}$  such that  $n_S \leq |S| + 1$ .
2: Compute  $\hat{\Sigma}_{\mathbb{S}} = \text{SampleCorr } X_{\mathbb{S}}$  and  $\hat{c} = \min_{S \in \mathbb{S}} \lambda_{\min}(\hat{\Sigma}_S)$ .
3: Compute  $R(\hat{\Sigma}_{\mathbb{S}})$  and the dual decomposition  $\hat{\Sigma}_{\mathbb{S}} = (1 - R(\hat{\Sigma}_{\mathbb{S}}))\hat{Q}_{\mathbb{S}} + R(\hat{\Sigma}_{\mathbb{S}})\hat{\Sigma}'_{\mathbb{S}}$ .
4: if  $R(\hat{\Sigma}_{\mathbb{S}}) \geq 3/4$  then
5:   return  $p_R = 0$ .
6: else
7:   For all  $S \in \mathbb{S}$ , for all  $i \in [n_S]$  do  $\tilde{X}_{S,i} := \hat{Q}_S^{1/2} \hat{\Sigma}_S^{-1/2} \text{diag}^{-1/2}(\hat{\sigma}_S^2)(X_{S,i} - \hat{\mu}_S)$ .
8:   for  $b \in [B]$  do
9:     For all  $S \in \mathbb{S}$ , let  $\tilde{X}_S^{(b)} = (\tilde{X}_{S,i}^{(b)} : i \in [n_S])$  be a nonparametric bootstrap sample from  $\tilde{X}_S$ .
10:    Compute  $\hat{\Sigma}_{\mathbb{S},b} = \text{SampleCorr } \tilde{X}_{\mathbb{S}}^{(b)}$ .
11:    Compute  $R_{\hat{c}/2}(\hat{\Sigma}_{\mathbb{S},b})$ .
12:   end for
13:   return  $p_R := (1 + B)^{-1} (1 + \sum_{b=1}^B \mathbb{1}\{R_{\hat{c}/2}(\hat{\Sigma}_{\mathbb{S},b}) \geq R(\hat{\Sigma}_{\mathbb{S}})\})$ .
14: end if

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for some sample size n_S and some distribution P_S with mean μ_S , correlation matrix Σ_S , vector of variances σ_S^2 and covariance matrix $\Omega_S = \text{diag}^{1/2}(\sigma_S^2) \cdot \Sigma_S \cdot \text{diag}^{1/2}(\sigma_S^2)$. We write $\hat{\mu}_S$, $\hat{\Sigma}_S$, $\hat{\sigma}_S^2$ and $\hat{\Omega}_S$ for the sample mean, the sample correlation matrix, the vector of sample variances and the sample covariance matrix, respectively, of $X_{S,1}, \dots, X_{S,n_S}$ for each $S \in \mathbb{S}$. Additionally, we write $\hat{\mu}_{\mathbb{S}} = (\hat{\mu}_S : S \in \mathbb{S})$, $\hat{\Sigma}_{\mathbb{S}} = (\hat{\Sigma}_S : S \in \mathbb{S})$, $\hat{\sigma}_{\mathbb{S}}^2 = (\hat{\sigma}_S^2 : S \in \mathbb{S})$ and $\hat{\Omega}_{\mathbb{S}} = (\hat{\Omega}_S : S \in \mathbb{S})$ for the collections of these estimators.

We now propose a bootstrap test of MCAR that can be applied without any knowledge of unknown parameters. This will be based on the incompatibility index $R(\cdot) \in [0, 1]$ defined in Section 3, which acts on correlation matrices and characterises compatibility, in the sense that $R(\Sigma_{\mathbb{S}}) = 0$ if and only if $\Sigma_{\mathbb{S}}$ is compatible, and its regularised version $R_z(\cdot)$ defined in Equation (7). Algorithm 1, which is implemented the R-package `MCARtest` (Berrett et al. (2022)), tackles the testing problem $H_0 : R(\Sigma_{\mathbb{S}}) = 0$ using as test statistic the plug-in estimate $R(\hat{\Sigma}_{\mathbb{S}})$, and is calibrated through a bootstrap procedure. As already outlined in Section 1, rejection of H_0 is sufficient to reject MCAR.

As far as the computational aspects of the algorithm are concerned, Proposition 5 in Section 3, together with equations (6) and (7), ensures that both $R(\hat{\Sigma}_{\mathbb{S}})$ and the regularised index $R_{\hat{c}/2}(\hat{\Sigma}_{\mathbb{S},b})$ can be characterised as the optimal value of a SDP problem. These programs can be efficiently solved in practice using standard optimisation libraries, for example, the R package `Rcsdp`. The proof of Proposition 5 shows how to express $R(\cdot)$ in standard SDP form. Moreover, the dual decomposition, specifically $\hat{Q}_{\mathbb{S}}$ and $\hat{\Sigma}'_{\mathbb{S}}$, corresponds to the optimal solution to the dual SDP problem. This quantity is also returned by standard solvers and can be extracted directly when computing $R(\hat{\Sigma}_{\mathbb{S}})$.

The intuition behind the procedure is as follows. From Proposition 5, we can write $\hat{\Sigma}_{\mathbb{S}} = (1 - R(\hat{\Sigma}_{\mathbb{S}}))\hat{Q}_{\mathbb{S}} + R(\hat{\Sigma}_{\mathbb{S}})\hat{\Sigma}'_{\mathbb{S}}$, where $\hat{Q}_{\mathbb{S}}$ can be thought as the closest compatible sequence of correlation matrices to $\hat{\Sigma}_{\mathbb{S}}$, and can be computed at the same time as the test statistic $R(\hat{\Sigma}_{\mathbb{S}})$, and $\hat{\Sigma}'_{\mathbb{S}}$ is an arbitrary sequence of correlation matrices. If $R(\hat{\Sigma}_{\mathbb{S}}) \geq 3/4$, we reject the null hypothesis outright, as there is strong evidence against it. Otherwise, we proceed by calibrating our test using a bootstrap procedure. We transform the original data by calculating $\tilde{X}_S := \hat{Q}_S^{1/2} \hat{\Sigma}_S^{-1/2} \text{diag}^{-1/2}(\hat{\sigma}_S^2)(X_S - \hat{\mu}_S)$ for all $S \in \mathbb{S}$. This transformation means that the sample correlation matrices of $\tilde{X}_{\mathbb{S}} := (\tilde{X}_S : S \in \mathbb{S})$ are given by $\hat{Q}_{\mathbb{S}}$, which is compatible. Fixing $B \in \mathbb{N}_+$, for each $b \in [B]$ and $S \in \mathbb{S}$ we generate $\tilde{X}_S^{(b)}$ as a nonparametric bootstrap sample from $\tilde{X}_S$ and calculate the sample correlation matrix $\hat{\Sigma}_{S,b} = \text{SampleCorr}(\tilde{X}_S^{(b)})$. Then, for each $b \in [B]$ we compute the corresponding test statistic $R_{\hat{c}/2}(\hat{\Sigma}_{S,b} : S \in \mathbb{S})$. These

will serve as surrogates of the null, and will be employed to generate a p -value of the form $p_R := (1 + B)^{-1} (1 + \sum_{b=1}^B \mathbb{1}\{R_{\hat{c}/2}(\hat{\Sigma}_{\mathbb{S},b}) \geq R(\hat{\Sigma}_{\mathbb{S}})\})$. The reason why we are forced to use the regularised version $R_{\hat{c}/2}(\cdot)$ at the bootstrap level instead of the standard $R(\cdot)$, and why we treat the case $R(\hat{\Sigma}_{\mathbb{S}}) \geq 3/4$ separately, is that it is notably challenging to characterise the spectrum of $\hat{Q}_{\mathbb{S}}$. This approach might inflate the size of our test in general. However, this inflation is vanishingly small since $R_{\hat{c}/2}(\hat{\Sigma}_{\mathbb{S},1}) \mid X_{\mathbb{S}} \xrightarrow{\mathbb{P}} R(\hat{\Sigma}_{\mathbb{S},1}) \mid X_{\mathbb{S}}$, as demonstrated, for instance, in the proof of Proposition 2.

The theoretical properties of Algorithm 1 are based on analysing the concentration properties of the plug-in estimator $R(\hat{\Sigma}_{\mathbb{S}})$ and the bootstrap statistics $R_{\hat{c}/2}(\hat{\Sigma}_{\mathbb{S},b})$, which require several technical insights. These are developed in Section 4. In light of the results presented therein, we are able to prove that our bootstrap test is uniformly valid over expanding subsets of the null, and uniformly powerful over alternatives separated from the null. To this end, let

$$\bar{\mathcal{P}}_{\mathbb{S}}(0) := \{P_{\mathbb{S}} \in \mathcal{P}_{\mathbb{S}} : R(\Sigma_{\mathbb{S}}) = 0\}.$$

For $P_{\mathbb{S}} \in \mathcal{P}_{\mathbb{S}}$ and $\epsilon > 0$, define

$$B_{\epsilon}(P_{\mathbb{S}}) = \left\{P'_{\mathbb{S}} \in \mathcal{P}_{\mathbb{S}} : \max_{S \in \mathbb{S}} \|\Sigma_S - \Sigma'_S\|_2 \leq \epsilon\right\} \quad \text{and} \\ \bar{\mathcal{P}}_{\mathbb{S}}(0)^{-\epsilon} = \{P_{\mathbb{S}} \in \bar{\mathcal{P}}_{\mathbb{S}}(0) : B_{\epsilon}(P_{\mathbb{S}}) \subseteq \bar{\mathcal{P}}_{\mathbb{S}}(0)\}.$$

THEOREM 1. *Suppose we observe $X_{S,1}, \dots, X_{S,n_S} \stackrel{i.i.d.}{\sim} P_S$ for each $S \in \mathbb{S}$ independently, where each P_S is ν -sub-Gaussian with mean μ_S . Denote by $\Sigma_{\mathbb{S}}$ the collection of population correlation matrices and by $\Omega_{\mathbb{S}}$ the collection of population covariance matrices, and assume that $\Sigma_{\mathbb{S}} \succeq_{\mathbb{S}} cI_{\mathbb{S}}$ for $c > 0$. Let $\sigma_{\min}^2 := \min_{j \in [d]} \min_{S \in \mathbb{S}_j} (\Omega_S)_{jj}$ and define*

$$(1) \quad C_{\alpha,c} = \frac{K_1 \nu^4}{c \sigma_{\min}^4} \max_{S \in \mathbb{S}} \sqrt{\frac{|S| + \log(|\mathbb{S}|/\alpha)}{n_S}}.$$

(i) *Then, for a universal constant $K_1 > 0$ chosen sufficiently large, for all $\alpha \in (0, 1)$, if $C_{\alpha,2} \leq 1$, we have*

$$\sup_{P_{\mathbb{S}} \in \bar{\mathcal{P}}_{\mathbb{S}}(0)^{-C_{\alpha,2}}} \mathbb{P}_{P_{\mathbb{S}}} \{p_R \leq \alpha\} \leq \alpha.$$

(ii) *Moreover, if $\alpha \in (0, 1)$ and $\beta \in (0, 1 - \alpha)$ are such that $B \geq 2(1 - \alpha)/\alpha$ and $\tilde{C}_{\alpha,\beta} \leq 1/2$, and if $P_{\mathbb{S}} \in \mathcal{P}_{\mathbb{S}}$ is such that $R(\Sigma_{\mathbb{S}}) > 2\tilde{C}_{\alpha,\beta}$, then we have $\mathbb{P}_{P_{\mathbb{S}}} \{p_R > \alpha\} \leq \beta$, where*

$$\tilde{C}_{\alpha,\beta} = \max \left(\max_{S \in \mathbb{S}} \frac{K_2 \nu^5}{\sigma_{\min}^5 c^{3/2}} \sqrt{\frac{\log(n_S) \log(|\mathbb{S}|/\alpha\beta) \{|S| + \log(|\mathbb{S}|n_S/\alpha\beta)\}}{n_S}}, \right. \\ \left. \max_{S \in \mathbb{S}} \frac{K_3 \nu^6}{\sigma_{\min}^6 c^2} \frac{\{|S| + \log(|\mathbb{S}|/\alpha\beta)\} \log^2(|\mathbb{S}|n_S/\alpha\beta)}{n_S}, C_{\alpha\beta,c} \right),$$

and $K_2, K_3 > 0$ are sufficiently large universal constants.

First, observe that the initial part of the result implies asymptotic validity across the entire null hypothesis, except at its boundary. Additionally, compared to the oracle test given in Theorem 7 in Section 4, the separation required to achieve uniform power includes just an

extra logarithmic term in n_S and $|\mathbb{S}|$, being of the order

$$\tilde{C}_{\alpha,\beta} \lesssim \max_{S \in \mathbb{S}} \sqrt{\frac{|S| \log n_S \log(|\mathbb{S}|/\alpha\beta)}{n_S}},$$

when c , $\sigma_{\min}^2$ and v^2 are fixed. This, combined with Theorems 10 and 12, also demonstrates that our bootstrap test is essentially minimax optimal up to logarithmic factors when $\mathbb{S}$ is either a d -cycle or a block 3-cycle.

Regarding the behaviour at the boundary of the null hypothesis, it is important to note that classical bootstrap procedures can be inconsistent for certain parameter values on the boundary of the associated parameter spaces (Andrews (2000), Samworth (2003), Cavaliere, Nielsen and Rahbek (2017)). While alternative approaches like the m out of n bootstrap can restore consistency in these cases, they may still be outperformed by the inconsistent standard bootstrap in practice. For example, this phenomenon is highlighted in Samworth (2003) through an analysis of the parametric bootstrap in the context of the Hodges and Stein estimators. As for the bootstrap test given in Algorithm 1, its asymptotic validity on the boundary crucially depends on the geometrical properties of the null hypothesis, which are in turn induced by the structure of the collection of missingness patterns $\mathbb{S}$. While this is generally very complex, we can still demonstrate the asymptotic validity of the bootstrap test over the entire null hypothesis in certain specific cases, such as for the d -cycle.

To this end, recall that for bivariate random vectors $(X_1, Y_1), \dots, (X_n, Y_n) \stackrel{\text{i.i.d.}}{\sim} P$ with mean (ξ, η) , variances (σ^2, τ^2) , and finite fourth moments, we have that $\sqrt{n}(\hat{\rho} - \rho) \xrightarrow{d} N(0, \gamma^2(P))$ (Lehmann (1999), Example 5.4.3), where ρ and $\hat{\rho}$ are Pearson correlation and its sample version, respectively. Here, the definition of $\gamma^2(P)$ is as follows: let $(S)_{ij} = \sigma_{ij}$ with

$$\begin{aligned} \sigma_{11} &= \text{Var}[(X - \xi)(Y - \eta)] = E[(X - \xi)^2(Y - \eta)^2 - \rho^2\sigma^2\tau^2], \\ \sigma_{12} &= \sigma_{21} = \text{Cov}[(X - \xi)(Y - \eta), (X - \xi)^2] = E[(X - \xi)^3(Y - \eta) - \rho\sigma^3\tau], \\ \sigma_{22} &= \text{Var}(X - \xi)^2 = E(X - \xi)^4 - \sigma^4, \\ \sigma_{13} &= \sigma_{31} = \text{Cov}[(X - \xi)(Y - \eta), (Y - \eta)^2] = E[(X - \xi)(Y - \eta)^3] - \rho\sigma\tau^3, \\ \sigma_{23} &= \sigma_{32} = \text{Cov}[(X - \xi)^2, (Y - \eta)^2] = E[(X - \xi)^2(Y - \eta)^2] - \sigma^2\tau^2, \\ \sigma_{33} &= \text{Var}(Y - \eta)^2 = E(Y - \eta)^4 - \tau^4. \end{aligned} \tag{2}$$

Let $f(u, v, w) = u/\sqrt{vw}$, and define $\gamma^2(P) = a^T S a$, where $a = (\partial f/\partial u, \partial f/\partial v, \partial f/\partial w)$ evaluated at $u = \rho\sigma\tau$, $v = \sigma^2$, $w = \tau^2$.

PROPOSITION 2. *Consider a d -cycle, that is, $\mathbb{S} = \{\{1, 2\}, \{2, 3\}, \dots, \{d, 1\}\}$, and write $\Sigma_{\mathbb{S}} := (\Sigma_{\{1,2\}}, \dots, \Sigma_{\{d,1\}})$ for the collection of 2×2 correlation matrices with*

$$\Sigma_{j,j+1} = \begin{pmatrix} 1 & \cos \theta_j \\ \cos \theta_j & 1 \end{pmatrix},$$

where $\theta_j \in [0, \pi]$ for all $j \in [d]$. Let $P_{\{j,j+1\}}$ be the distribution of $(X_j, X_{j+1}) \mid \Omega = \mathbb{1}_{\{j,j+1\}}$, and suppose $P_{\{j,j+1\}}$ has finite fourth moments for all $j \in [d]$. Assume that $\min\{1 + \cos \theta_j, 1 - \cos \theta_j\} \geq c$ for all $j \in [d]$ for a universal constant $c \in (0, 1)$, and suppose there exists $j \in [d]$ such that $\gamma^2(P_{\{j,j+1\}}) > 0$. For all $P_{\mathbb{S}} \in \bar{\mathcal{P}}_{\mathbb{S}}(0)$ and $\alpha < 1/2$, we have

$$\lim_{n_{\mathbb{S}} \rightarrow \infty} \lim_{B \rightarrow \infty} \mathbb{P}_{P_{\mathbb{S}}} \{p_R \leq \alpha\} \leq \alpha.$$

Furthermore, if $P_{\mathbb{S}} \in \partial \bar{\mathcal{P}}_{\mathbb{S}}(0) := \bar{\mathcal{P}}_{\mathbb{S}}(0) \setminus \bigcup_{\epsilon > 0} \bar{\mathcal{P}}_{\mathbb{S}}(0)^{-\epsilon}$, the probability of rejection tends exactly to α .

The condition $\gamma^2(P_{j,j+1}) > 0$ ensures that $R(\hat{\Sigma}_{\mathbb{S}})$ and $R_{\hat{c}/2}(\hat{\Sigma}_{\mathbb{S},b})$ do not have a degenerate limit, and holds for all sufficiently regular distributions. For instance, in the case of elliptical distributions, we have $\gamma^2(P) = (1 + \kappa)(1 - \rho^2)^2 > 0$ (see Theorem 5.1.6 in [Muirhead \(1982\)](#)), since $\rho^2 < 1$ by assumption, and the kurtosis κ is guaranteed to satisfy $\kappa > -1$ ([Bentler and Berkane \(1986\)](#)).

3. Measure of incompatibility for covariance matrices. In this section, we develop our index $R(\cdot)$ of the incompatibility of population covariance matrices, which will be defined as the optimal value of a semidefinite programme. Standardising the covariance matrices is necessary to have a well-posed problem, and we choose to work with correlation matrices because this is most natural from a statistical point of view. Other standardisation are possible, though, each leading to a different measure of incompatibility, with different properties. In the Supplementary Material A ([Bordino and Berrett \(2025\)](#)), we introduce another measure of incompatibility based on a different standardisation, analyse its properties and derive an oracle test of MCAR based on its estimation from data. In order to define and study the properties of $R(\cdot)$, we must first introduce some basic algebraic objects for collections of symmetric and positive semidefinite matrices. Our key notation is collected in Table 1 below.

Crucially, we say that an element of $\mathcal{P}_{\mathbb{S}}^*$ is *compatible* if and only if it is an element of $\mathcal{P}_{\mathbb{S}}^{0,*}$. It turns out that we can characterise compatibility through the adjoint of the linear operator A defined in Table 1, which maps a symmetric matrix X into a collection of symmetric matrices indexed by $\mathbb{S}$ according to $(AX)_S = (X_{jj'})_{j,j' \in S}$ for all $S \in \mathbb{S}$.

PROPOSITION 3. *The adjoint operator $A^* : \mathcal{M}_{\mathbb{S}} \rightarrow \mathcal{M}$ of A is given by*

$$(A^*X_{\mathbb{S}})_{jj'} = \sum_{S \in \mathbb{S}} \mathbb{1}_{\{j,j' \in S\}} (X_S)_{jj'} = \sum_{S \in \mathbb{S}_{jj'}} (X_S)_{jj'},$$

where we recall that $\mathbb{S}_{jj'} = \{S \in \mathbb{S} : j, j' \in S\} = \mathbb{S}_j \cap \mathbb{S}_{j'}$.

TABLE 1
Table of definitions commonly used in the main text

Notation	Definition	Meaning
$\mathbb{S}$	A subset of the power set of $[d]$	Set of all missingness patterns
$\mathbb{S}_j$	$\{S \in \mathbb{S} : j \in S\}$	Set of all patterns that contain j
$\mathbb{S}_{jj'}$	$\{S \in \mathbb{S} : j, j' \in S\}$	Set of all patterns that contain (j, j')
$\mathcal{M} \equiv \mathcal{M}_d$	$\{X \in \mathbb{R}^{d \times d} : X = X^T\}$	Space of symmetric matrices
$\mathcal{M}_{\mathbb{S}}$	$\{(X_S : S \in \mathbb{S}) : X_S \in \mathcal{M}_{ S } \text{ for all } S \in \mathbb{S}\}$	Space of collections of symmetric matrices
$\langle X, Y \rangle$	$\text{tr}(XY)$ for $X, Y \in \mathcal{M}$	Frobenius inner product
$\langle X_{\mathbb{S}}, Y_{\mathbb{S}} \rangle_{\mathbb{S}}$	$\sum_{S \in \mathbb{S}} \text{tr}(X_S Y_S)$ for $X_{\mathbb{S}}, Y_{\mathbb{S}} \in \mathcal{M}_{\mathbb{S}}$	Sum of Frobenius inner products
$\mathcal{P}^*$	$\{\Sigma \in \mathcal{M} : \Sigma \succeq 0\}$	Cone of PSD matrices
$\mathcal{P}$	$\{\Sigma \in \mathcal{P}^* : \text{diag}(\Sigma) = I_d\}$	Set of correlation matrices
$\Sigma_{\mathbb{S}} \succeq_{\mathbb{S}} 0$	$\Sigma_{\mathbb{S}} \succeq_{\mathbb{S}} 0$ if and only if $\Sigma_S \succeq 0$ for all $S \in \mathbb{S}$	Loewner order for collections of matrices
$\mathcal{P}_{\mathbb{S}}^*$	$\{\Sigma_{\mathbb{S}} \in \mathcal{M}_{\mathbb{S}} : \Sigma_{\mathbb{S}} \succeq_{\mathbb{S}} 0\}$	Set of collections of PSD matrices
$\mathcal{P}_{\mathbb{S}}$	$\{\Sigma_{\mathbb{S}} \in \mathcal{P}_{\mathbb{S}}^* : \text{diag}(\Sigma_S) = I_{ S } \text{ for all } S \in \mathbb{S}\}$	Set of collections of correlation matrices
A	$A : \mathcal{M} \rightarrow \mathcal{M}_{\mathbb{S}}$ with $(AX)_S = (X_{jj'})_{j,j' \in S}$	Marginalisation operator on matrices
$\mathcal{P}_{\mathbb{S}}^{0,*}$	$\{A\Sigma : \Sigma \in \mathcal{P}^*\}$	Set of compatible collections of PSD matrices
$\mathcal{P}_{\mathbb{S}}^0$	$\{A\Sigma : \Sigma \in \mathcal{P}\}$	Set of compatible collections of correlation matrices
$\mathcal{Y}$	$\{\text{diag}(v) : v \in \mathbb{R}^d \text{ and } \sum_{j=1}^d v_j = 0\}$	Space of diagonal matrices with null trace
$\mathbf{0}_{\mathbb{S}}$	$(\mathbf{0}_{ S } : S \in \mathbb{S})$	Collection of zero matrices
$I_{\mathbb{S}}$	$(I_{ S } : S \in \mathbb{S})$	Collection of identity matrices

Now, the following proposition fully characterises compatibility in terms of the non-negativity of a collection of linear functionals.

PROPOSITION 4. *For $\Sigma_{\mathbb{S}} \in \mathcal{P}_{\mathbb{S}}^*$, we have $\Sigma_{\mathbb{S}} \in \mathcal{P}_{\mathbb{S}}^{0,*}$ if and only if*

$$\langle X_{\mathbb{S}}, \Sigma_{\mathbb{S}} \rangle_{\mathbb{S}} \geq 0 \quad \text{for all } X_{\mathbb{S}} \in \mathcal{M}_{\mathbb{S}} \text{ satisfying } A^* X_{\mathbb{S}} \geq 0.$$

This is an extension of well-known characterisation of the feasibility of positive semi-definite matrix completion (e.g., [Laurent \(2009\)](#)). Indeed, when $\Sigma_{\mathbb{S}}$ is consistent, we can show that $\langle X_{\mathbb{S}}, \Sigma_{\mathbb{S}} \rangle_{\mathbb{S}} = \langle A^* X_{\mathbb{S}}, \Sigma \rangle$, where Σ is the $d \times d$ symmetric matrix with $\Sigma_{jj'} = (\Sigma_S)_{jj'}$ for all $S \in \mathbb{S}_{jj'}$, if $\mathbb{S}_{jj'} \neq \emptyset$, and $\Sigma_{jj'} = 0$ if $\mathbb{S}_{jj'} = \emptyset$. Here, Σ can be thought of as a partial matrix that is padded with zeros in unobserved positions. Since $A^* X_{\mathbb{S}}$ is also zero in these positions, the value of Σ there is arbitrary. Now our characterisation reduces to checking that $\langle A^* X_{\mathbb{S}}, \Sigma \rangle \geq 0$ for all $X_{\mathbb{S}} \in \mathcal{M}_{\mathbb{S}}$ satisfying $A^* X_{\mathbb{S}} \geq 0$, which is equivalent to checking $\langle X, \Sigma \rangle \geq 0$ for all $X \in \mathcal{M}$ with $X_{jj'} = 0$ if $\mathbb{S}_{jj'} = \emptyset$, which coincides with (4) in [Laurent \(2009\)](#).

Proposition 4 provides a characterisation of compatibility, but in order to assess the significance of departures from the null hypothesis, and thus to define hypothesis tests, we will need a numerical measure of incompatibility. A natural way to do this is to minimise $\langle X_{\mathbb{S}}, \Sigma_{\mathbb{S}} \rangle_{\mathbb{S}}$ over a subset of $\{X_{\mathbb{S}} \in \mathcal{M}_{\mathbb{S}} : A^* X_{\mathbb{S}} \geq 0\}$ that still characterises compatibility, but gives finite minimal values. First, observe that checking the compatibility of covariance matrices is equivalent to checking the consistency of the variances $\sigma_S^2 = \text{diag}(\text{Cov}(X_S | \Omega = \mathbb{I}_S))$ for $S \in \mathbb{S}$, and the compatibility of the correlation matrices $\text{Corr}(X_S | \Omega = \mathbb{I}_S)$ for $S \in \mathbb{S}$. Here, we will focus on the latter issue, and postpone the discussion of testing the consistency of variances to Section 6. This is because this problem essentially reduces to testing the equality of variances, which is well understood in the statistical literature ([Brown and Forsythe \(1974\)](#), [Gastwirth, Gel and Miao \(2009\)](#)). Now, whenever $\Sigma_{\mathbb{S}}$ is a collection of correlation matrices we define

$$(3) \quad \begin{aligned} R(\Sigma_{\mathbb{S}}) &:= \sup \left\{ -\frac{1}{d} \langle \Sigma_{\mathbb{S}}, X_{\mathbb{S}} \rangle_{\mathbb{S}} : X_{\mathbb{S}} + X_{\mathbb{S}}^0 \succeq_{\mathbb{S}} 0, A^* X_{\mathbb{S}} + Y \succeq 0 \text{ for some } Y \in \mathcal{Y} \right\} \\ &= 1 - \frac{1}{d} \inf \{ \langle \Sigma_{\mathbb{S}}, Y_{\mathbb{S}} \rangle_{\mathbb{S}} : Y_{\mathbb{S}} \succeq_{\mathbb{S}} 0, A^* Y_{\mathbb{S}} + Y \succeq I_d \text{ for some } Y \in \mathcal{Y} \}, \end{aligned}$$

where $X_{\mathbb{S}}^0 = (X_S^0 : S \in \mathbb{S})$ with $X_S^0 = \text{diag}(1/|\mathbb{S}_j| : j \in S)$, and where $\mathcal{Y}$ is the set of diagonal $d \times d$ matrices with trace equal to zero. We refer to $R(\cdot)$ as an index of incompatibility, borrowing the terminology from [Berrett and Samworth \(2023\)](#). The objective function of this optimisation problem is a one-to-one mapping of the linear functional appearing in our characterisation of compatibility. Moreover, by choosing $Y = \mathbf{O}$ and noting that $X_{\mathbb{S}}^0 \succ_{\mathbb{S}} 0$, we can see that for any $X_{\mathbb{S}}$ that satisfies $A^* X_{\mathbb{S}} \geq 0$, the collection $\epsilon X_{\mathbb{S}}$ is feasible for $\epsilon > 0$ sufficiently small. Thus, by Proposition 4 we have that $R(\Sigma_{\mathbb{S}}) > 0$ whenever $\Sigma_{\mathbb{S}}$ is incompatible. On the other hand, when $\Sigma_{\mathbb{S}} = A\Sigma$ is compatible and $(X_{\mathbb{S}}, Y)$ is feasible we have $\langle \Sigma_{\mathbb{S}}, X_{\mathbb{S}} \rangle_{\mathbb{S}} = \langle \Sigma, A^* X_{\mathbb{S}} \rangle = \langle \Sigma, A^* X_{\mathbb{S}} + Y \rangle \geq 0$, where the second equality holds because Σ has a constant diagonal. Combining this with the observation that $X_{\mathbb{S}} = \mathbf{O}_{\mathbb{S}}$ is feasible, we see that $R(\Sigma_{\mathbb{S}}) = 0$ when $\Sigma_{\mathbb{S}}$ is compatible.

In the above argument, we did not use the specific form of the lower bound $X_{\mathbb{S}} \succeq_{\mathbb{S}} -X_{\mathbb{S}}^0$ anywhere, and it would also have been possible to optimise over the restricted set of $X_{\mathbb{S}}$ that are feasible with $Y = \mathbf{O}$. The specific choice of the feasible set in the definition of $R(\Sigma_{\mathbb{S}})$ was made because it leads to an interpretable dual formulation. While there exist semidefinite programs for which strong duality does not hold, Slater's condition is satisfied in our problem, so we do not encounter such issues (see the Supplementary Material A ([Bordino and Berrett \(2025\)](#)) for an introduction to the theory of semidefinite programming). This is formalised in the result below.

PROPOSITION 5. For $\Sigma_{\mathbb{S}} \in \mathcal{P}_{\mathbb{S}}$, we have

$$(4) \quad \begin{aligned} R(\Sigma_{\mathbb{S}}) &= \inf\{\epsilon \in [0, 1] : \Sigma_{\mathbb{S}} \in (1 - \epsilon)\mathcal{P}_{\mathbb{S}}^0 + \epsilon\mathcal{P}_{\mathbb{S}}\} \\ &= 1 - \frac{1}{d} \sup\{\text{tr}(\Sigma) : A\Sigma \preceq_{\mathbb{S}} \Sigma_{\mathbb{S}}, \Sigma_{11} = \dots = \Sigma_{dd}, \Sigma \succeq 0\}. \end{aligned}$$

This result shows that our measure of incompatibility $R(\Sigma_{\mathbb{S}})$ can be interpreted as the smallest amount of perturbation ϵ that a compatible collection of correlation matrices must be corrupted by to result in the input collection $\Sigma_{\mathbb{S}}$. It is immediate from this representation that $R(\Sigma_{\mathbb{S}})$ takes values in $[0, 1]$. Moreover, it follows from Slater's condition that the optimal value of the dual problem is attained. Thus, writing $\lambda^* = 1 - R(\Sigma_{\mathbb{S}})$, we can write $\Sigma_{\mathbb{S}}$ as

$$(5) \quad \Sigma_{\mathbb{S}} = \lambda^* A\Sigma + (1 - \lambda^*)\Sigma'_{\mathbb{S}},$$

where $\Sigma \in \mathcal{P}$ and $\Sigma'_{\mathbb{S}} \in \mathcal{P}_{\mathbb{S}}$. By the maximality of $\lambda^* = 1 - R(\Sigma_{\mathbb{S}})$, it must be the case that $R(\Sigma'_{\mathbb{S}}) = 1$. Indeed, if this were not the case, it would be possible to write $\Sigma'_{\mathbb{S}} = \lambda' A\Sigma' + (1 - \lambda')\Sigma''_{\mathbb{S}}$ for some $\lambda' \in (0, 1]$, which would contradict the fact that λ^* is optimal. This argument shows that, whenever $\mathbb{S}$ is such that there exists an incompatible collection $\Sigma_{\mathbb{S}}$, the maximal value $R(\Sigma_{\mathbb{S}}) = 1$ is attainable, so that the quantity $R(\Sigma_{\mathbb{S}})$ is on an interpretable scale between compatibility at one extreme and maximal incompatibility at the other.

We remark that this dual interpretation of $R(\cdot)$ aligns with a similar representation of the incompatibility of collections of distributions defined by [Berrett and Samworth \(2023\)](#). In this earlier work it is shown that the incompatibility of collections of distributions can be understood through linear programming techniques. Our work here, however, shows that we must consider the more complex problem of semidefinite programming to understand the incompatibility of collections of covariance matrices. Despite this additional complexity, since Slater's condition is satisfied for our problem, the primal-dual interior point method has a computational complexity which is polynomial in the number of constraints and the dimension of the variable square matrix (Section 6.4.1. of [Nesterov and Nemirovskii \(1994\)](#), Section 5.7. of [Vandenberghe and Boyd \(1996\)](#)). This ensures that $R(\Sigma_{\mathbb{S}})$ can be always computed efficiently without additional assumptions.

We conclude this section with some basic properties of $R(\cdot)$ that will be used in later proofs.

PROPOSITION 6. The following hold:

- (i) $R(\cdot)$ is convex.
- (ii) $R(\cdot)$ is continuous.
- (iii) Suppose $\mathbb{S} \subseteq \mathbb{S}'$ and $\Sigma_{\mathbb{S}} \subseteq_{\mathbb{S}} \Sigma_{\mathbb{S}'}$, where the inclusion $\subseteq_{\mathbb{S}}$ means that every correlation matrix in $\Sigma_{\mathbb{S}}$ is also in $\Sigma_{\mathbb{S}'}$. Then $R(\Sigma_{\mathbb{S}}) \leq R(\Sigma_{\mathbb{S}'})$.

It is interesting to observe that property (iii) says that R is monotone with respect to the inclusion operator, so that additional information can only make a collection less compatible.

4. Concentration bound and an oracle test. Having introduced our population-level measure of incompatibility, in this section we analyse the concentration properties of the plug-in estimator $R(\hat{\Sigma}_{\mathbb{S}})$. This will give us tools that enable us to prove theoretical guarantees (Theorem 1) for the bootstrap procedures defined in Section 2. Furthermore, it will lead naturally to the definition of the oracle test defined in Theorem 7. Now, the analysis of $R(\hat{\Sigma}_{\mathbb{S}})$ is challenging, as it is defined as the optimal value of a semidefinite program with an unbounded feasible set. In fact, without further assumptions, it is not possible to restrict

attention to a compact feasible set. On the other hand, most statistical techniques for the analysis of suprema of empirical processes require feasible sets to be totally bounded so that, for example, covering arguments can be applied.

Fortunately, under the assumption that $\Sigma_{\mathbb{S}} \succ_{\mathbb{S}} 0$, our dual problem (4) is strictly feasible, and hence Slater's condition implies that the optimal value is attained in the primal problem (3). This assumption is reasonable in many areas of application, and similar assumptions of invertibility have been used frequently in the literature (Meinshausen and Bühlmann (2006), Cai, Liu and Luo (2011)). In fact, if we assume the stronger condition that $\Sigma_{\mathbb{S}} \succeq_{\mathbb{S}} cI_{\mathbb{S}}$ for some $c > 0$, we will see that the optimal value is always attained in a compact set whose size depends on c . Indeed, the strict feasibility of the dual problem (4) implies that there exists $X_{\mathbb{S}} \in \mathcal{M}_{\mathbb{S}}$ such that $X_{\mathbb{S}} + X_{\mathbb{S}}^0 \succeq_{\mathbb{S}} 0$ and $R(\Sigma_{\mathbb{S}}) = -d^{-1}\langle X_{\mathbb{S}}, \Sigma_{\mathbb{S}} \rangle_{\mathbb{S}}$. This in turn implies that

$$\sum_{S \in \mathbb{S}} \|X_S + X_S^0\|_* = \langle X_{\mathbb{S}} + X_{\mathbb{S}}^0, I_{\mathbb{S}} \rangle_{\mathbb{S}} \leq \frac{1}{c} \langle X_{\mathbb{S}} + X_{\mathbb{S}}^0, \Sigma_{\mathbb{S}} \rangle_{\mathbb{S}} = \frac{d}{c} \{1 - R(\Sigma_{\mathbb{S}})\} \leq \frac{d}{c},$$

so that we have a bound on the sum of the nuclear norms of the matrices in the sequence $X_{\mathbb{S}} + X_{\mathbb{S}}^0$. In finding the optimal value of the primal problem (3), then we may restrict attention to

$$(6) \quad \mathcal{F}_c := \left\{ X_{\mathbb{S}} \in \mathcal{M}_{\mathbb{S}} : X_{\mathbb{S}} + X_{\mathbb{S}}^0 \succeq_{\mathbb{S}} 0, \sum_{S \in \mathbb{S}} \|X_S + X_S^0\|_* \leq d/c, A^* X_{\mathbb{S}} + Y \succeq 0 \text{ for some } Y \in \mathcal{Y} \right\},$$

which is compact. Formally, defining the regularised version of $R(\cdot)$ as

$$(7) \quad R_z(\Sigma_{\mathbb{S}}) := \sup \{ -d^{-1} \langle X_{\mathbb{S}}, \Sigma_{\mathbb{S}} \rangle_{\mathbb{S}} : X_{\mathbb{S}} \in \mathcal{F}_z \} \quad \text{for } z \leq 1,$$

we have just shown that $R_c(\Sigma_{\mathbb{S}}) = R(\Sigma_{\mathbb{S}})$ whenever $\Sigma_{\mathbb{S}} \succeq_{\mathbb{S}} cI_{\mathbb{S}}$. This regularised index of incompatibility is used to compute the bootstrap statistic $R_{\hat{c}/2}(\hat{\Sigma}_{\mathbb{S},1})$ in Algorithm 1, and is such that $R_{z_1}(\Sigma_{\mathbb{S}}) \geq R_{z_2}(\Sigma_{\mathbb{S}})$ if $z_2 \geq z_1$.

Before moving on to describe how to construct a statistical test under this new assumption, we give a brief discussion of the norm on $\mathcal{M}_{\mathbb{S}}$ defined by

$$\|X_{\mathbb{S}}\|_{*,\mathbb{S}} := \sum_{S \in \mathbb{S}} \|X_S\|_*,$$

which reduces to $\sum_{S \in \mathbb{S}} \text{tr}(X_S)$ in case that $X_{\mathbb{S}} \succeq_{\mathbb{S}} 0$. For each $S \in \mathbb{S}$, the nuclear norm $\|X_S\|_*$ can be thought of as the ℓ_1 norm applied to the eigenvalues of X_S . As these are then summed to give $\|X_{\mathbb{S}}\|_{*,\mathbb{S}}$, we see that $\|\cdot\|_{*,\mathbb{S}}$ can be thought of as an ℓ_1 norm on $\mathcal{M}_{\mathbb{S}}$. It is easy to see that the dual norm of $\|\cdot\|_{*,\mathbb{S}}$ with respect to the inner product $\langle \cdot, \cdot \rangle_{\mathbb{S}}$ is

$$\|X_{\mathbb{S}}\|_{2,\mathbb{S}} := \max_{S \in \mathbb{S}} \|X_S\|_2,$$

where $\|\cdot\|_2$ is the usual spectral norm of a matrix. This follows after writing the sequence of matrices in block-diagonal form, and allows us to derive the following generalisation of Holder's inequality in the space of matrix collections:

$$(8) \quad |\langle X_{\mathbb{S}}, Y_{\mathbb{S}} \rangle_{\mathbb{S}}| \leq \|X_{\mathbb{S}}\|_{*,\mathbb{S}} \|Y_{\mathbb{S}}\|_{2,\mathbb{S}} = \sum_{S \in \mathbb{S}} \|Y_S\|_* \cdot \max_{S \in \mathbb{S}} \|X_S\|_2.$$

This inequality will be used in the proof of the following result, which provides valid critical values for the test statistic $R(\hat{\Sigma}_{\mathbb{S}})$ and gives conditions under which the resulting test has large power.

THEOREM 7. *Suppose that the assumptions of Theorem 1 hold, and recall the definition of $C_{\alpha,c} \equiv C_\alpha$ from (1). Then, for a universal constants $K_1 > 0$ chosen sufficiently large, for all $\alpha \in (0, 1)$, the test that rejects $H_0 : R(\Sigma_{\mathbb{S}}) = 0$ if and only if $R(\hat{\Sigma}_{\mathbb{S}}) \geq C_\alpha$ has Type-I error bounded by α . Moreover, for $\beta \in (0, 1 - \alpha)$, if $R(\Sigma_{\mathbb{S}}) > C_\alpha + C_\beta$, then $\mathbb{P}\{R(\hat{\Sigma}_{\mathbb{S}}) \leq C_\alpha\} \leq \beta$.*

In proving this result, we give concentration inequalities for the random quantities $R(\hat{\Sigma}_{\mathbb{S}})$. The analysis of $R(\hat{\Sigma}_{\mathbb{S}})$ is crucially based on the fact that, under H_0 and in light of the inequality (8), we can control the oscillation $|R(\hat{\Sigma}_{\mathbb{S}}) - R(\Sigma_{\mathbb{S}})|$ using $\max_{S \in \mathbb{S}} \|\hat{\Sigma}_S - \Sigma_S\|_2$, where the Σ_S are the Pearson population correlation matrices and $\hat{\Sigma}_S$ are the corresponding Pearson sample correlation matrices. To this end, we derive a tail bound for the spectral norm $\|\hat{\Sigma} - \Sigma\|_2$, where Σ is the population correlation matrix and $\hat{\Sigma}$ is the sample correlation matrix of complete data, which may be of independent interest. This can be found in the Supplementary Material A (Bordino and Berrett (2025)).

As well as providing a critical value for our test, Theorem 7 also gives upper bounds on the minimax separation rate for this testing problem. When $c > 0$, $\sigma_{\min}^2$ and $\nu > 0$ are fixed, our analysis gives an upper bound on the minimax rate of the order

$$C_\alpha \lesssim \max_{S \in \mathbb{S}} \sqrt{\frac{|S| + \log(|\mathbb{S}|/\alpha)}{n_S}},$$

whenever $n_S \gtrsim |S|$ for all $S \in \mathbb{S}$. This is our main regime of interest, and we see in our examples in Section 5 below that reliable testing is only possible when sample sizes are large compared with dimensions, up to logarithmic factors.

We conclude this section by illustrating the behaviour of this bound in certain examples where the expression for C_α can be simplified. The corresponding upper bounds on the minimax separation rate will be complemented by lower bounds in Section 5 to follow.

EXAMPLE 1. In the d -cycle example, with $d \geq 3$ and $\mathbb{S} = \{\{1, 2\}, \{2, 3\}, \dots, \{d-1, d\}, \{d, 1\}\}$, we have $|S| = 2$ for all $S \in \mathbb{S}$ and $|\mathbb{S}| = d$ so that

$$C_\alpha \lesssim \max_{S \in \mathbb{S}} \sqrt{\frac{\log(d/\alpha)}{n_S}} = \sqrt{\frac{\log(d/\alpha)}{\min_{S \in \mathbb{S}} n_S}}.$$

Combined with Theorem 10 below, this reveals that, in this specific example, testing the compatibility of the correlation matrices is no harder than testing the consistency of the variances, up to constant factors.

EXAMPLE 2. Consider the block-3-cycle $\mathbb{S} = \{[2d], [d] \cup ([3d] \setminus [2d]), [3d] \setminus [d]\}$, with $d \geq 1$. Then

$$C_\alpha \lesssim \max_{S \in \mathbb{S}} \sqrt{\frac{d + \log(1/\alpha)}{n_S}} = \sqrt{\frac{d + \log(1/\alpha)}{\min_{S \in \mathbb{S}} n_S}}.$$

We prove the minimax optimality, up to logarithmic factors, of this rate in Theorem 12. In particular, this shows that the optimal separation rates for this testing problem are not significantly faster than the optimal rates for the estimation of $\Sigma_{\mathbb{S}}$ with operator norm loss.

EXAMPLE 3. Consider $\mathbb{S} = \text{Pow}([d])$, where $\text{Pow}(\cdot)$ stands for the power set. This corresponds to the case where we observe all possible missingness patterns from a dataset of dimension d . In this case, we have $\max_{S \in \mathbb{S}} |S| = d$ and $\log |\mathbb{S}| = d \log 2$, so that

$$C_\alpha \lesssim \max_{S \in \mathbb{S}} \sqrt{\frac{d + \log(1/\alpha)}{n_S}} = \sqrt{\frac{d + \log(1/\alpha)}{\min_{S \in \mathbb{S}} n_S}}.$$

This shows that C_α is at worst of the order $\sqrt{d/n}$.

5. Optimality and examples. In this section, we assess the optimality of the oracle test given in Theorem 7, in the settings of Examples 1 and 2, that is, when $\mathbb{S} = \{\{1, 2\}, \{2, 3\}, \dots, \{d-1, d\}, \{d, 1\}\}$, the d -cycle, and when $\mathbb{S} = \{[2d], [d] \cup ([3d] \setminus [2d]), [3d] \setminus [d]\}$, the block 3-cycle. These collections $\mathbb{S}$ provide examples where our methodology is provably near rate-optimal. For a given dimension d , these two examples further demonstrate the range of optimal rates that can arise for different collections $\mathbb{S}$. Assuming for simplicity that $n_S = n$ for all $S \in \mathbb{S}$, we will see that the optimal rate in the d -cycle case is $\{\log(d)/n\}^{1/2}$, while for the block 3-cycle it is $(d/n)^{1/2}$ up to logarithmic factors. Together, these results show that the structure of $\mathbb{S}$ can have a significant effect on the difficulty of the problem.

We will characterise the optimality of a testing procedure using the minimax framework, where we aim at finding the smallest separation between the null and the alternative hypotheses such that there exists a test that can distinguish between H_0 and H_1 up to a given level of error. More precisely, given $\rho > 0$, we are interested in testing

$$H_0 : R(\Sigma_{\mathbb{S}}) = 0 \quad \text{vs.} \quad H_1 : R(\Sigma_{\mathbb{S}}) > \rho,$$

and our goal is to find the smallest value of ρ such that there exists a test with uniform error control. Write $\Psi \equiv \Psi_{\mathbb{S}}(n_{\mathbb{S}})$ for the set of all tests, that is measurable functions of the data $(X_{S,i} : S \in \mathbb{S}, i \in [n_S])$ taking values in $\{0, 1\}$. Recall that $\tilde{\mathcal{P}}_{\mathbb{S}}(0)$ denotes the set of all collections of distributions on $(\mathbb{R}^S : S \in \mathbb{S})$ such that the associated correlation matrices satisfy $R(\Sigma_{\mathbb{S}}) = 0$, and write $\mathcal{P}_{\mathbb{S}}(\rho)$ for the set of all collections of distributions on $(\mathbb{R}^S : S \in \mathbb{S})$ such that the associated correlation matrices satisfy $R(\Sigma_{\mathbb{S}}) > \rho$. Given a collection of distributions $P_{\mathbb{S}} = (P_S : S \in \mathbb{S})$ on $(\mathbb{R}^S : S \in \mathbb{S})$ and a collection of sample sizes $n_{\mathbb{S}} = (n_S : S \in \mathbb{S})$, we write $P_{\mathbb{S}}^{\otimes n_{\mathbb{S}}}$ for the distribution of the entire dataset $(X_{S,i} : S \in \mathbb{S}, i \in [n_S])$ when each observation is independent and $X_{S,i} \sim P_S$ for each $i \in [n_S]$ and $S \in \mathbb{S}$. For a fixed $\eta \in (0, 1)$, we may then define the minimax separation to be

$$\rho^* \equiv \rho_{\mathbb{S}}^*(n_{\mathbb{S}}, \eta) := \inf \left\{ \rho > 0 : \inf_{\varphi \in \Psi} \left(\sup_{P_{\mathbb{S},0} \in \tilde{\mathcal{P}}_{\mathbb{S}}(0)} P_{\mathbb{S},0}^{\otimes n_{\mathbb{S}}}(\varphi = 1) + \sup_{P_{\mathbb{S},1} \in \mathcal{P}_{\mathbb{S}}(\rho)} P_{\mathbb{S},1}^{\otimes n_{\mathbb{S}}}(\varphi = 0) \right) \leq \eta \right\}.$$

In our analysis, we take $\eta = 3/4$, but this is an arbitrary choice and any constant value in $(0, 1)$ would result in the same qualitative behaviour. In common with previous work on minimax testing, we prove lower bounds on ρ^* by constructing suitable (prior) distributions μ_0, μ_1 whose support is contained in $\tilde{\mathcal{P}}_{\mathbb{S}}(0), \mathcal{P}_{\mathbb{S}}(\rho)$, respectively. In our proofs, it will be sufficient to consider mean-zero Gaussian distributions with suitable priors over their covariance matrices. Having chosen these priors, we can bound the minimal error probability by writing

$$\begin{aligned} \sup_{P_{\mathbb{S},0} \in \tilde{\mathcal{P}}_{\mathbb{S}}(0)} P_{\mathbb{S},0}^{\otimes n_{\mathbb{S}}}(\varphi = 1) + \sup_{P_{\mathbb{S},1} \in \mathcal{P}_{\mathbb{S}}(\rho)} P_{\mathbb{S},1}^{\otimes n_{\mathbb{S}}}(\varphi = 0) &\geq \mathbb{E}_{\mu_0} P_{\mathbb{S},0}^{\otimes n_{\mathbb{S}}}(\varphi = 1) + \mathbb{E}_{\mu_1} P_{\mathbb{S},1}^{\otimes n_{\mathbb{S}}}(\varphi = 0) \\ &\geq 1 - \text{TV}(\mathbb{E}_{\mu_0} P_{\mathbb{S},0}^{\otimes n_{\mathbb{S}}}, \mathbb{E}_{\mu_1} P_{\mathbb{S},1}^{\otimes n_{\mathbb{S}}}), \end{aligned}$$

where $\mathbb{E}_{\mu_i} P_{\mathbb{S},i}^{\otimes n_{\mathbb{S}}}$ denotes the mixture distribution of the dataset resulting from generating $P_{\mathbb{S},i} \sim \mu_i$ then, conditionally on $P_{\mathbb{S},i}$, generating the data. Then the idea behind this method of finding a lower bound on ρ^* is to find priors μ_0, μ_1 with the largest separation possible such that no test can successfully distinguish between $\mathbb{E}_{\mu_0} P_{\mathbb{S},0}^{\otimes n_{\mathbb{S}}}$ and $\mathbb{E}_{\mu_1} P_{\mathbb{S},1}^{\otimes n_{\mathbb{S}}}$.

5.1. Cycles. Recall that we refer to $\mathbb{S}$ as a d -cycle when $\mathbb{S} = \mathbb{S}_d := \{\{1, 2\}, \dots, \{d, 1\}\}$. In this subsection, additions in subscripts of the form $(j, j+1)$ for $j \in [d]$ are intended modulo d , where d is the size of the cycle, so that $(0, 1)$ and $(d, d+1)$ are equivalent to $(d, 1)$. We also write $\Sigma_{\mathbb{S}_d} := (\Sigma_{\{1,2\}}, \dots, \Sigma_{\{d,1\}})$ for a collection of 2×2 correlation matrices with

$$\Sigma_{j,j+1} = \begin{pmatrix} 1 & \cos \theta_j \\ \cos \theta_j & 1 \end{pmatrix},$$

and $\theta_j \in [0, \pi]$ for all $j \in [d]$.

Now, recall that Theorem 7 implies an upper bound on the minimax separation of the form $\rho^* \lesssim \sqrt{\log d/n}$. While it is straightforward to show a lower bound of $\rho^* \gtrsim 1/\sqrt{n}$ using a standard Le Cam two-point argument, matching the logarithmic dependence on d in the numerator is technically challenging. Nonetheless, we will argue that $\sqrt{\log d/n}$ is the optimal rate of convergence in the combined problem where we test both the compatibility of the correlation matrices and the consistency of the variances.

To this end, we now define an analogous index of inconsistency for the collection of variances. Writing $\sigma_{\mathbb{S}}^2 = (\sigma_S^2 : S \in \mathbb{S})$ for the collection of individual variances, we fix our units of measurement such that

$$\bar{a}v_j(\sigma_{\mathbb{S}}^2) := |\mathbb{S}_j|^{-1} \sum_{S \in \mathbb{S}_j} \sigma_{S,j}^2 = 1,$$

for all $j \in [d]$, where $\sigma_{S,j}^2$ is the j th element of σ_S^2 . This is a natural constraint, analogous to the standardisation of variables in complete-data problems, that does not remove information that may be present in the individual variances. For such $\sigma_{\mathbb{S}}^2$, define

$$V(\sigma_{\mathbb{S}}^2) := 1 - \min_{j \in [d]} \min_{S \in \mathbb{S}_j} \sigma_{S,j}^2 = \max_{j \in [d]} \max_{S \in \mathbb{S}_j} (1 - \sigma_{S,j}^2).$$

Under the hypothesis $\bar{a}v_j(\sigma_{\mathbb{S}}^2) = 1$ for all $j \in [d]$, it is clear that $V(\sigma_{\mathbb{S}}^2) = 0$ if and only if $\sigma_{S,j}^2 = 1$ for all $j \in [d]$, $S \in \mathbb{S}_j$. On the other hand, we have $V(\sigma_{\mathbb{S}}^2) > 0$, if and only if there exists at least one variance strictly less than 1. It is clear from the definition that V is bounded by one, and that this extreme value is attainable when $\mathbb{S}$ is nontrivial and there exists j such $\sigma_{S,j}^2 = 0$ for some $S \in \mathbb{S}_j$. The following result gives a dual representation for $V(\sigma_{\mathbb{S}}^2)$, providing justification for our specific measure of inconsistency.

PROPOSITION 8. *If $\bar{a}v_j(\sigma_{\mathbb{S}}^2) = 1$ for all $j \in [d]$, then*

$$V(\sigma_{\mathbb{S}}^2) = \inf\{\epsilon \in [0, 1] : \sigma_{\mathbb{S}}^2 = (1 - \epsilon)A_V \mathbf{1}_d + \epsilon \sigma_{\mathbb{S}}'^2 \text{ with } \bar{a}v_j(\sigma_{\mathbb{S}}'^2) = 1 \text{ for all } j \in [d]\},$$

where $(A_V \sigma^2)_S = (\sigma_k^2)_{k \in S}$.

This result gives a dual representation for $V(\sigma_{\mathbb{S}}^2)$, which is analogous to Proposition 5 and leads to a similar interpretation, based on the idea of finding the smallest perturbation to make the collection consistent.

We now characterise the optimal rate of convergence in the combined problem where we test both the compatibility of the correlation matrices and the consistency of the variances. Together, the following results show that the oracle test based on the plug-in estimator $R(\hat{\Sigma}_{\mathbb{S}}) + V(\hat{\sigma}_{\mathbb{S}}^2)$ is optimal up to constant factors, and further imply that testing the consistency of the variances captures the essential statistical difficulty of the problem in the case of a d -cycle.

THEOREM 9. *Suppose that the assumptions of Theorem 1 hold, and further assume that the sequence of variances $\sigma_{\mathbb{S}}^2$ satisfies $\bar{a}v_j(\sigma_{\mathbb{S}}^2) = 1$ for all $j \in [d]$. Define $C_{\alpha}^{(R+V)} = \max\{2C_{\alpha/2}, C_{\alpha}^{(V)}\}$, with $C_{\alpha} \equiv C_{\alpha,c}$ as in (1) and*

$$C_{\alpha}^{(V)} = K_2 v^2 \sqrt{\frac{\log(\sum_{S \in \mathbb{S}} |S|/\alpha)}{\min_{S \in \mathbb{S}} n_S}},$$

for a universal constant $K_2 > 0$. Then, for $K_1, K_2 > 0$ chosen sufficiently large, for all $\alpha \in (0, 1)$, if $C_{\alpha}^{(R+V)} \leq \min\{2, 16v^2\}$, the test that rejects $H_0 : R(\Sigma_{\mathbb{S}}) + V(\sigma_{\mathbb{S}}^2) = 0$ if and only if $R(\hat{\Sigma}_{\mathbb{S}}) + V(\hat{\sigma}_{\mathbb{S}}^2) \geq C_{\alpha}^{(R+V)}$ has Type-I error bounded by α . Moreover, for all $\beta \in (0, 1 - \alpha)$, if

$C_\beta^{(R+V)} \leq \min\{2, 16v^2\}$ and $R(\Sigma_{\mathbb{S}}) + V(\sigma_{\mathbb{S}}^2) > C_\alpha^{(R+V)} + C_\beta^{(R+V)}$, then $\mathbb{P}\{R(\hat{\Sigma}_{\mathbb{S}}) + V(\hat{\sigma}_{\mathbb{S}}^2) \leq C_\alpha^{(R+V)}\} \leq \beta$.

THEOREM 10. Let $\mathbb{S} = \mathbb{S}_d$ for $d \geq 3$, with sample sizes $n_{\mathbb{S}} = (n_1, \dots, n_d)$, and consider testing

$$H_0' : R(\Sigma_{\mathbb{S}}) + V(\sigma_{\mathbb{S}}^2) = 0 \quad \text{vs.} \quad H_1' : R(\Sigma_{\mathbb{S}}) + V(\sigma_{\mathbb{S}}^2) > \rho,$$

for $\rho > 0$. Call ρ_{R+V}^* the minimax separation of this testing problem. There exists a universal constant $c_1 > 0$ such that

$$\rho_{R+V}^* \geq c_1 \sqrt{\frac{\log d}{\min_{j \in [d]} n_j}}.$$

Furthermore, we can give a relatively explicit expression for $R(\cdot)$ for a general d -cycle.

PROPOSITION 11. Let Σ be the optimum solution to the dual problem (4), and let $\varphi^* = (\varphi_1^*, \dots, \varphi_d^*)$ be such that $\Sigma_{j,j+1} = \cos \varphi_j^*$ for each $j \in [d]$. Then:

- (i) $R(\Sigma_{\mathbb{S}_d}) = |\rho_j - (1 - R(\Sigma_{\mathbb{S}_d})) \cos \varphi_j^*|$, for all $j \in [d]$;
- (ii) $\varphi^* = (\varphi_1^*, \dots, \varphi_d^*)$ is unique, and $\varphi^*(\theta_1, \dots, \theta_d)$ is continuous for varying $(\theta_1, \dots, \theta_d) \in [0, \pi]^d$;
- (iii) if $\theta_1 = \max_{j \in [d]} \theta_j$, with $\theta_2, \dots, \theta_d \leq \pi/2$, then

$$1 - R(\Sigma_{\mathbb{S}_d}) = \frac{1 - \epsilon_j \cos \theta_j}{1 - \epsilon_j \cos \varphi_j^*} \quad \text{for all } j \in [d],$$

where $\epsilon_d = (\epsilon_1, \dots, \epsilon_d) = (-1, +\mathbf{1}_{d-1})$. Also, $\varphi_1^* = \sum_{j=2}^d \varphi_j^*$.

Observe that part (ii) only says that the entries of Σ corresponding to the cycle pattern are unique, not the whole Σ itself. Indeed, given the unique optimal φ^* , there may exist infinitely many positive semidefinite completions. In fact, if A is a partial symmetric matrix admitting a positive semidefinite completion, then there exists a unique positive semidefinite completion with maximum determinant (Grone et al. (1984), Theorem 2). For a general sequence of angles $(\theta_1, \dots, \theta_d)$, it is sufficient to use the transformation given in Proposition S6 in the Supplementary Material A (Bordino and Berrett (2025)) to reduce to the case where at most one angle is larger than $\pi/2$, choose ϵ_d as outlined above, and perform the inverse transformation to obtain the signs for the original $(\theta_1, \dots, \theta_d)$. As an immediate corollary of this, it is easy to see that, under the same set of hypotheses, we have

$$1 - R(\Sigma_{\mathbb{S}_d}) = \frac{1 + \cos \theta_1}{1 + \cos \varphi_1^*},$$

where φ_1^* is the solution of

$$\begin{cases} \varphi_1^* = \sum_{j=2}^d \varphi_j^*, \\ \cos \varphi_j^* = 1 - \frac{1 - \cos \theta_j}{1 + \cos \theta_1} (1 + \cos \varphi_1^*) \quad \text{for all } j \in \{2, \dots, d\}. \end{cases}$$

For further properties of $R(\cdot)$ in the case of a d -cycle, we refer the reader to the Supplementary Material A (Bordino and Berrett (2025)).

5.2. *Block cycles.* So far, we have studied with particular care the case of a d -cycle, which is a relatively simple high-dimensional setting, since it is a collection of d two-dimensional distributions. We now describe an evolution of this setting, where we consider a block-matrix version of the 3-cycle. In this case, the number of variables per missingness pattern is large and we will see that the minimax separation rates are correspondingly much larger than in the d -cycle, though the number of variables is of the same order.

THEOREM 12. *Let $\mathbb{S} = \{[2d], [d] \cup ([3d] \setminus [2d]), [3d] \setminus [d]\}$ for some $d \geq 1$. Writing $n_{\mathbb{S}} = (n_1, n_2, n_3)$ for the sample sizes within each pattern, there exists a universal constant $c_1 > 0$ such that*

$$\rho^* \geq c_1 \sqrt{\frac{d}{(n_1 \wedge n_2) \log^4(ed)}}$$

whenever $n_1 \wedge n_2 \geq d/2$.

This result shows that, up to logarithmic factors in d , the minimax separation rates for this testing problem are the same as the minimax estimation rates for estimating $\Sigma_{\mathbb{S}}$ in the operator norm distance. This is related to the fact that $R(\Sigma_{\mathbb{S}})$ is a nonsmooth functional of $\Sigma_{\mathbb{S}}$. Indeed, the following result shows that we can construct examples of $\Sigma_{\mathbb{S}}$ such that $R(\Sigma_{\mathbb{S}})$ can be bounded below using the function $x \mapsto \max(0, x)$; see below for more discussion of the relevant literature.

PROPOSITION 13. *Consider the set of patterns $\mathbb{S} = \{[2d], [d] \cup ([3d] \setminus [2d]), [3d] \setminus [d]\}$ for some $d \geq 1$, and suppose that*

$$\Sigma_{\mathbb{S}} = \left\{ \begin{pmatrix} I_d & P \\ P^T & I_d \end{pmatrix}, \begin{pmatrix} I_d & -P \\ -P^T & I_d \end{pmatrix}, \begin{pmatrix} I_d & \beta I_d \\ \beta I_d & I_d \end{pmatrix} \right\},$$

for some $P \in \mathbb{R}^{d \times d}$ such that $\|P\|_2 \leq 1$ and some $\beta \in [0, 1]$. Then:

- (i) $R(\Sigma_{\mathbb{S}}) = 0$ if and only if $\|P\|_2^2 \leq \frac{1-\beta}{2}$,
- (ii) $R(\Sigma_{\mathbb{S}}) \geq \frac{3}{4d} \sum_{j=1}^d (\sigma_j^2(P) - \frac{1-\beta}{2})_+$, where $\sigma_j(P)$ is the j th singular value of P .

This shows that, for $\Sigma_{\mathbb{S}}$ of the form above, we can relate our testing problem to the problem of testing whether the vector of squared singular values of P belongs to the orthant $(-\infty, (1 - \beta)/2]^d$, or is separated from it in the ℓ_1 distance. In a Gaussian location model a similar problem, measuring separation with the ℓ_2 distance, was considered by Blanchard, Carpentier and Gutzeit (2018), and part of our lower bound construction is inspired by this work. However, the consideration of singular values of matrices rather than Gaussian means means that new technical tools are required. In this regard, the techniques of Thépaut and Verzelen (2024), who consider the estimation of quantities of the form $\sum_{j=1}^d \sigma_j(P)^q$ for $q > 0$, are useful. We also mention that such problem are related to the estimation of ℓ_1 distances, for which good references include Cai and Low (2011) and Jiao, Han and Weissman (2018).

6. **Simulations.** In this section, we empirically validate the performance of the bootstrap test described in Algorithm 1. To also detect departures from the null hypothesis caused by inconsistencies in either the means or variances, we introduce two separate bootstrap procedures addressing these aspects individually. Specifically, Algorithm 2, which provides the p -value p_M , is designed to detect inconsistencies in the collection of means $\mu_{\mathbb{S}}$, while Algorithm 3 returns the p -value p_V , focusing on inconsistencies in the variances $\sigma_{\mathbb{S}}^2$. To

Algorithm 2 Means only: MCAR bootstrap test checking consistency of means

-
- 1: Given data $X_{\mathbb{S}}$, discard all patterns $S \in \mathbb{S}$ such that $n_S \leq |S| + 10$.
 - 2: Compute $\hat{\mu}_{\mathbb{S}} = \text{SampleMean } X_{\mathbb{S}}$, i.e., $\hat{\mu}_{S,j} = n_S^{-1} \sum_{i \in n_S} X_{S,ij}$ for all $S \in \mathbb{S}$ and $j \in S$.
 - 3: Compute $M(\hat{\mu}_{\mathbb{S}}) = \max_{S \in \mathbb{S}} \|\hat{\mu}_S - \hat{\mu}_{|S}\|_1 / \max_{S \in \mathbb{S}} |S|$, where $(\hat{\mu})_j = \hat{\mu}_j = |S_j|^{-1} \sum_{S \in \mathbb{S}_j} \hat{\mu}_{S,j}$.
 - 4: Rotate the original data $X_{\mathbb{S}}$, i.e., for all $S \in \mathbb{S}$, for all $i \in [n_S]$ **do** $\tilde{X}_{S,i} = X_{S,i} - \hat{\mu}_S + \hat{\mu}_{|S}$.
 - 5: **for** $b \in [B]$ **do**
 - 6: For all $S \in \mathbb{S}$, let $\tilde{X}_S^{(b)} = (\tilde{X}_{S,i}^{(b)} : i \in [n_S])$ be a nonparametric bootstrap sample from $\tilde{X}_S$.
 - 7: Compute $\hat{\mu}_{\mathbb{S},b} = \text{SampleMean } X_{\mathbb{S}}^{(b)}$ and $M(\hat{\mu}_{\mathbb{S},b})$.
 - 8: **end for**
 - 9: **return** $p_M := (1 + B)^{-1} (1 + \sum_{b=1}^B \mathbb{1}\{M(\hat{\mu}_{\mathbb{S},b}) \geq M(\hat{\mu}_{\mathbb{S}})\})$.
-

create a more comprehensive test, we propose an *omnibus* procedure that combines these three p -values using a Bonferroni correction, which consists in rejecting the null whenever $\min\{p_R, p_V, p_M\} < \alpha/3$. Although alternative methods for combining p -values, such as Fisher's method (Mosteller and Fisher (1948)), could be considered, we chose this approach for its simplicity and its better control of the Type-I error. Furthermore, there are various alternative approaches for measuring inconsistency in means and variances, but we chose these because they align with the spirit of Algorithm 1. Additionally, this is a relatively classical problem since it reduces to testing the equality of means and variances. Consequently, alternative methods might achieve better practical performance. However, this is beyond the scope of our work, as the real novelty lies in addressing the more challenging problem of testing the compatibility of correlation matrices.

We will compare Algorithm 1 (represented by the purple line in the plots) and the *omnibus* approach (represented by the blue line in the plots) with Little's test (Little (1988)). Little's test can be applied when all pairs of variables are observed together, so that the EM algorithm (Dempster, Laird and Rubin (1977)) can be applied to find estimators $\hat{\mu}$ and $\hat{\Lambda}$ of the mean and covariance matrix of the data under the null hypothesis of MCAR. Little's test is a generalised likelihood ratio test whose validity is based on the assumption that the data $(X_{S,i} : S \in \mathbb{S}, i \in [n_S])$ are Gaussian. Define

$$d_{\mu}^2 = \sum_{S \in \mathbb{S}} n_S (\bar{X}_S - \hat{\mu}_{|S}) \Lambda_{|S}^{-1} (\bar{X}_S - \hat{\mu}_{|S}),$$

$$d_{\text{cov}}^2 = \sum_{S \in \mathbb{S}} n_S [\text{tr}(\hat{\Omega}_S \hat{\Lambda}_{|S}^{-1}) - |S| - \log |\hat{\Omega}_S| + \log |\hat{\Lambda}_{|S}|],$$

Algorithm 3 Variance only: MCAR bootstrap test checking consistency of variances

-
- 1: Given data $X_{\mathbb{S}}$, discard all patterns $S \in \mathbb{S}$ such that $n_S \leq |S| + 10$.
 - 2: Compute $\hat{\sigma}_{\mathbb{S}}^2 = \text{SampleVar } X_{\mathbb{S}}$, i.e., $\hat{\sigma}_{S,j}^2 = n_S^{-1} \sum_{i \in n_S} X_{S,ij}^2 - \hat{\mu}_{S,j}^2$ for all $S \in \mathbb{S}$ and $j \in S$.
 - 3: Rescale the data such that $|S_j|^{-1} \sum_{S \in \mathbb{S}_j} \hat{\sigma}_{S,j}^2 = 1$ for all $j \in [d]$.
 - 4: Compute $V(\hat{\sigma}_{\mathbb{S}}^2) = 1 - \min_{S \in \mathbb{S}} \min_{j \in S} \hat{\sigma}_{S,j}^2$.
 - 5: Rotate the original data $X_{\mathbb{S}}$, i.e., for all $S \in \mathbb{S}$, for all $i \in [n_S]$ **do** $\tilde{X}_{S,i} = \text{diag}(\hat{\sigma}_S^{-1}) X_{S,i}$.
 - 6: **for** $b \in [B]$ **do**
 - 7: For all $S \in \mathbb{S}$, let $\tilde{X}_S^{(b)} = (\tilde{X}_{S,i}^{(b)} : i \in [n_S])$ be a nonparametric bootstrap sample from $\tilde{X}_S$.
 - 8: Compute $\hat{\sigma}_{\mathbb{S},b}^2 = \text{SampleVar } \tilde{X}_{\mathbb{S},b}$ and rescale the data s.t. $|S_j|^{-1} \sum_{S \in \mathbb{S}_j} \hat{\sigma}_{S,b,j}^2 = 1$ for all $j \in [d]$.
 - 9: Compute $V(\hat{\sigma}_{\mathbb{S},b}^2)$.
 - 10: **end for**
 - 11: **return** $p_V := (1 + B)^{-1} (1 + \sum_{b=1}^B \mathbb{1}\{V(\hat{\sigma}_{\mathbb{S},b}^2) \geq V(\hat{\sigma}_{\mathbb{S}}^2)\})$.
-

and

$$d_{\text{aug}}^2 = d_{\mu}^2 + d_{\text{cov}}^2.$$

Then, under MCAR, d_{aug}^2 converges in law to a χ^2 -distribution with

$$f = \sum_{S \in \mathbb{S}} \frac{1}{2} |S| (|S| + 3) - \frac{1}{2} d (d + 3),$$

degrees of freedom by Wilks’ theorem. Based on these asymptotic results, Little’s test rejects MCAR if and only if $d_{\text{aug}}^2 > \chi_f^2(1 - \alpha)$, where $\chi_f^2(1 - \alpha)$ is such that $\mathbb{P}\{W \geq \chi_f^2(1 - \alpha)\} = \alpha$, and where W is χ^2 -distributed with f degrees of freedom. Using similar asymptotics, one can define a test based on d_{cov}^2 , which ignores the means and only considers the partial covariance matrices, and another one based only on d_{μ}^2 , which discards the collection of covariance matrices and makes use of the means only. For the test based on d_{μ}^2 , we will use the R-function `mcAR_test` from the R-package `nanian` (Tierney and Cook (2023)), while the other two tests based on d_{aug}^2 and d_{cov}^2 can be found in the R-package `MCARtest` (Berrett et al. (2022)) under the name `little_test`. In the following subsections, we will compare our procedures with Little’s test based on d_{μ}^2 (represented by the green line in the plots) and d_{aug}^2 (represented by the black line in the plots). We will also include a *combined* approach (represented by the orange line in the plots), which checks the compatibility of the covariance matrices using $\min\{p_R, p_V\}$ but checks the consistency of the means using d_{μ}^2 instead of Algorithm 2. The p -values are again combined using a Bonferroni correction. Section 6.1 is focused on the case where $\mathbb{S}$ is a d -cycle, which is the best-studied theoretical setting we considered in the paper, while Section 6.2 is closer to real-world data applications. The code to reproduce the simulations can be found in the Supplementary Material B. More details about the implementation of our procedures can be found in the documentation of the R-package `MCARtest`.

6.1. *Correlation matrices and simulations for d -cycles.* We compare Algorithm 1 with Little’s procedures in the settings given in Theorem 10, namely in the case of a d -cycle. For our first settings, we set $n_{\mathbb{S}} = (n_S)_{S \in \mathbb{S}} = (200, \dots, 200)$, and simulate $X_{\{j, j+1\}, i} \stackrel{\text{i.i.d.}}{\sim} N(\mathbf{0}_2, \Sigma_{\{j, j+1\}})$ for $i \in [200]$ and $j \in [d]$, where

$$\Sigma_{\mathbb{S}_d} = \left\{ \begin{pmatrix} 1 & \cos \theta_1 \\ \cos \theta_1 & 1 \end{pmatrix}, \dots, \begin{pmatrix} 1 & \cos \theta_d \\ \cos \theta_d & 1 \end{pmatrix} \right\},$$

for certain values of $\theta_1, \dots, \theta_d \in [0, \pi]$. This makes sense only for $d = 3$, while for $d \geq 4$ there exists at least one pair of variables that are never observed together, making the EM algorithm to estimate $\hat{\Lambda}$ inapplicable. As for the case $d = 3$, in Figure 2 we set $B = 99$ and $\alpha = 0.05$, and we vary $\theta_1 \in [\theta_2 + \theta_3, (\theta_2 + \theta_3 + \pi)/2]$, with (θ_2, θ_3) equal to $(\pi/3, \pi/6)$. We repeat the experiment $H = 500$ times, and report the average decision.

The simulation results depicted in Figure 2 show that for $d = 3$, both the *omnibus* and *combined* procedures perform similar to Little’s test based on d_{aug}^2 , though the Type-I error of Little’s test is slightly inflated. This outcome supports a conjecture in Little (1988), where it is suggested that even under normality, the asymptotic null distribution of d_{aug}^2 is unlikely to be reliable unless the sample size is large. Notably, Algorithm 1 demonstrates the highest power, which is not surprising since it is specifically designed to detect incompatible correlation matrices. As expected, the test based on d_{μ}^2 shows no power, as it is only sensitive to inconsistencies in the means, which are consistent in this particular scenario. For higher dimensions ($d \geq 4$), Little’s test cannot be applied, while our test remains valid since it has no constraints on $\mathbb{S}$. In Figure 3, we show the power function of our bootstrap tests in the case of

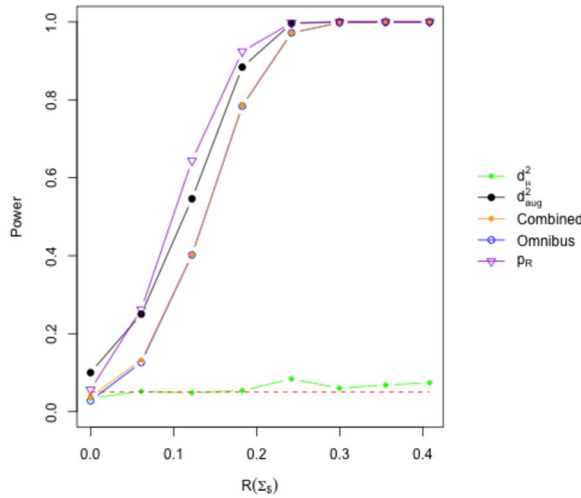


FIG. 2. 3-cycle with Gaussian data. We generate $X_{\{j,j+1\},i} \stackrel{i.i.d.}{\sim} N(\mathbf{0}_2, \Sigma_{\{j,j+1\}})$ for all $i \in [200]$, $j \in [3]$, where we vary $\theta_1 \in [\theta_2 + \theta_3, (\theta_2 + \theta_3 + \pi)/2]$, with $(\theta_2, \theta_3) = (\pi/3, \pi/6)$. We repeat the experiment $H = 500$ times, and report the average decision. The nominal level $\alpha = 0.05$ in red, $B = 99$.

a d -cycle, with $d \in \{100, 200\}$, with $\theta_2 = \dots = \theta_d = \frac{\pi}{2(d-1)}$, and varying θ_1 in $[\pi/2, 5\pi/8]$. We repeat the procedure $H = 500$ times, and report the average decision as an estimate of the power function.

Our simulations so far have used Gaussian data, so that Little’s test is valid. We now repeat our simulations with a heavy-tailed data distribution in order to assess the robustness of the methods. To this end, we consider again a 3-cycle, and generate $X_{\{j,j+1\},i} \stackrel{i.i.d.}{\sim} \log N(\mathbf{0}_2, \Sigma_{\{j,j+1\}})$ for all $i \in [200]$, $j \in [3]$, where $\log N(\mathbf{0}_2, \Sigma_{\{j,j+1\}})$ stands for the log-normal distribution, meaning that if $Y \sim \log N(\mathbf{0}_2, \Sigma_{\{j,j+1\}})$ then $Y_i = e^{Z_i}$, with $Z \sim N(\mathbf{0}_2, \Sigma_{\{j,j+1\}})$. Here, we vary $\theta_1 \in [\pi/2 + \pi/12, \pi/12]$, with $(\theta_2, \theta_3) = (3\pi/4, \pi/4)$. Figure 4 shows the analogue of Figure 2, in the case of artificial data from a multivariate log-

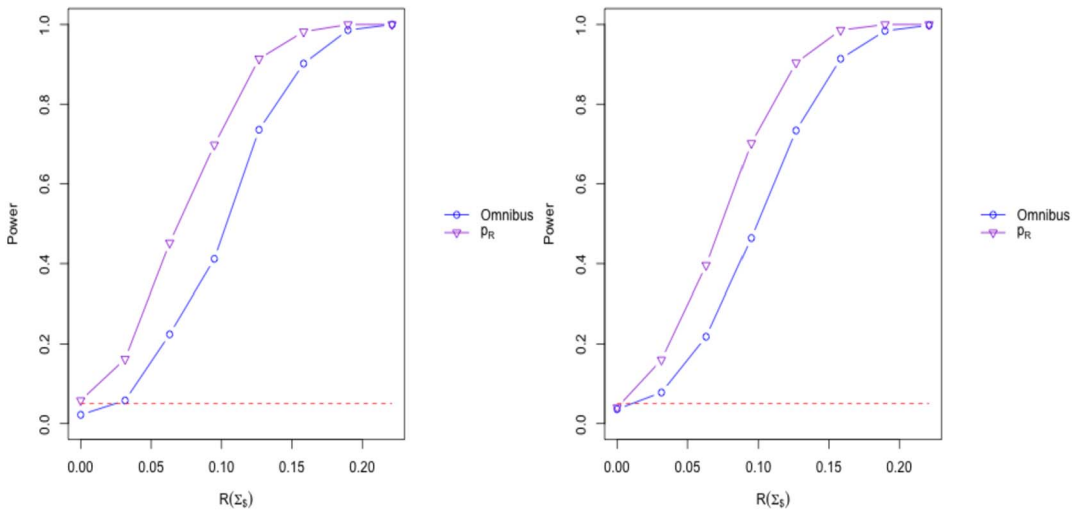


FIG. 3. Simulation of the power functions of our method with $B = 99$ (blue) for $d = 100$ (left), and $d = 200$ (right), with Gaussian data. In each example, we fix $\theta_2 = \dots = \theta_d = \frac{\pi}{2(d-1)}$, and vary θ_1 in $[\pi/2, 5\pi/8]$. For each of this setting, we repeat the experiment $H = 500$ times, and report the average decision. The nominal level $\alpha = 0.05$ in red.

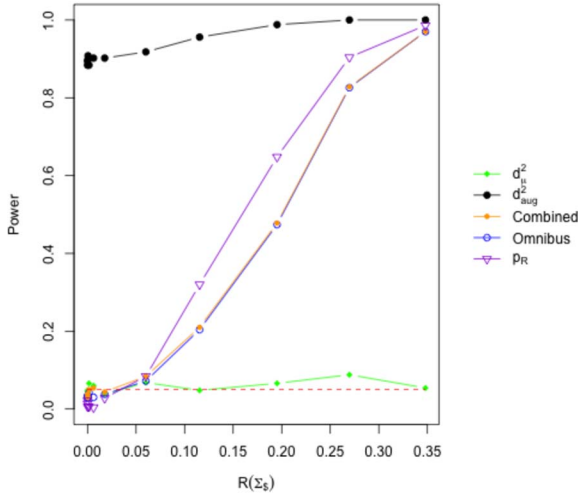


FIG. 4. 3-cycle with lognormal data. We generate $X_{\{j,j+1\},i} \stackrel{i.i.d.}{\sim} \log N(\mathbf{0}_2, \Sigma_{\{j,j+1\}})$ for all $i \in [200]$, $j \in [3]$, where we vary $\theta_1 \in [\pi/2 + \pi/12, \pi/12]$, while fixing $(\theta_2, \theta_3) = (3\pi/4, \pi/4)$. We use again $H = 500$, $B = 99$ and $\alpha = 0.05$. $R(\Sigma_S)$ is estimated using an independent sample.

normal distribution rather than a Gaussian distribution. Note however that the covariance matrices do not coincide with the original $\Sigma_{\{j,j+1\}}$, hence on the x -axis we decided to estimate $R(\Sigma_S)$ using $R(\hat{\Sigma}_S)$ using an independent sample from the same log-normal distribution. As in our previous results, Little’s test based on d^2_μ exhibits no power but maintains Type-I error control, even when we deviate from the Gaussian setting, which aligns with a conjecture made in Little (1988). In contrast, Little’s test based on d^2_{aug} fails to control the Type-I error. On the other hand, our three tests exhibit similar behavior to that in the Gaussian setting, with only a slight decrease in power.

6.2. Omnibus approach. In this subsection, we compare the omnibus and the combined approaches with Little’s tests in settings, which are closer to real-data applications. In this regard, we generate complete artificial data according to various distributions, and then delete entries using the R package missMethods (Rockel (2020)). MCAR data are generated with the function delete_MCAR, where each entry of the data matrix is deleted independently of the others with probability $p \in (0, 1)$. Deviations from the null are generated by partitioning the columns in two groups, group A where the missing values are generated, and group B, which determines the missingness mechanism, with two different mechanisms being considered. First, delete_MAR_1_to_x sets threshold values, splits the rows into two further groups depending on whether columns in group B have values greater or smaller than the threshold, and deletes some entries in columns in group A in a such a way that the probability for a value to be missing in group A divided by the probability for a value to be missing in group B equals 1 divided by x , with x to be specified as an input parameter. Second, delete_MAR_rank deletes each entry in a column of group A with probability proportional to the rank of the same row in the corresponding column of group B. For further details on these functions, and other methods to generate MCAR, MAR, MNAR data, refer to Santos et al. (2019). These three functions were also chosen in the numerical analysis of a test of MCAR based on U-statistics in Aleksić (2024). Before discussing the simulation results, we note that in these settings, under the alternative, the collection of correlation matrices is only mildly incompatible (as verified by numerical inspection), while the inconsistency in

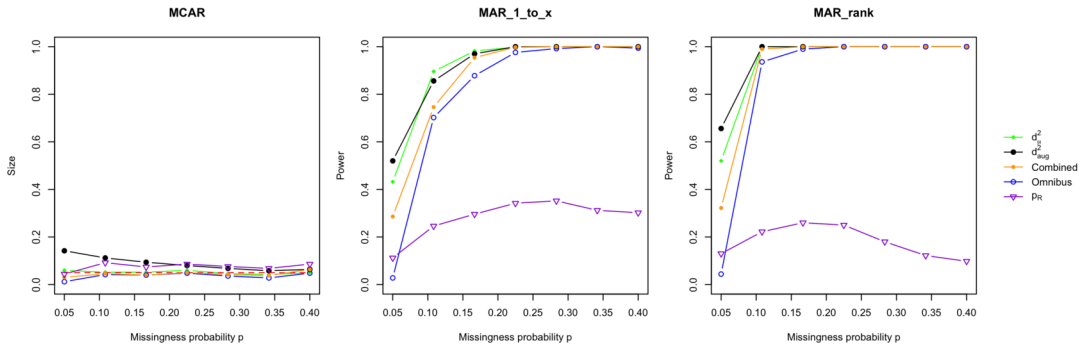


FIG. 5. (Left) Size for MCAR data generated with `delete_MCAR(p)`, for varying probability p of having a missing value; (Centre) Power function for MAR data generated with `delete_MAR_1_to_x(p, x = 9)`, for varying probability p of having a missing value; (Right) Power function for MAR data generated with `delete_MAR_rank(p)`, for varying probability p of having a missing value. For all three scenarios, we generate data from Clayton copula with parameter 1 and $N(0, 1)$ margins, applying our procedures with $B = 99$, $H = 500$.

means plays a critical role in detecting departures from MCAR. Consequently, the test based on p_R demonstrates low power, unlike our omnibus approach and Little's test based on d_μ^2 .

For Figure 5, we generated 3-dimensional datasets of sample size $n = 200$ distributed according to a Clayton copula, with parameter 1 and Gaussian ($N(0, 1)$) margins, using the function `mvdc` from the R-package `copula` (Hofert et al. (2020)). For the left panel, we deleted the first two variables with `delete_MCAR(p)` for different values of $p \in \{0.05, \dots, 0.40\}$, in order to get an artificial setting coming from the null. For each p , we repeat the simulation $H = 500$ times, and report the average Type-I error. Alternatives to the null were generated using `delete_MAR_1_to_x`, with $x = 9$, for the centre panel, and `delete_MAR_rank` for the right one. Again, for each p , we repeat the simulations 500 times, and report the average power. In this setting, simulation results shown in the centre and right panels demonstrate that all tests, apart from the one based on p_R , perform similarly in terms of power. The behaviour of the test based on p_R can be attributed to the fact that, under the alternative, the collection of correlation matrices is only mildly incompatible, with numerical inspection suggesting that incompatibility decreases as p increases. Meanwhile, the left panel seems to indicate that all tests control the Type-I error at the nominal

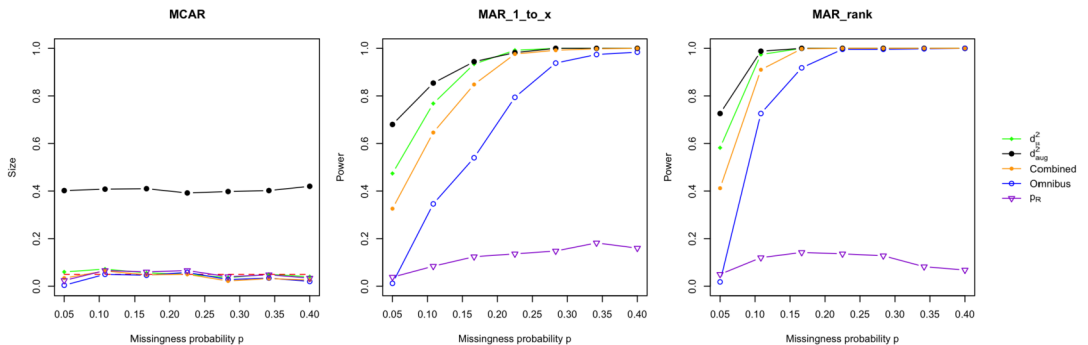


FIG. 6. (Left) Size for MCAR data generated with `delete_MCAR(p)`, for varying probability p of having a missing value; (Centre) Power function for MAR data generated with `delete_MAR_1_to_x(p, x = 9)`, for varying probability p of having a missing value; (Right) Power function for MAR data generated with `delete_MAR_rank(p)`, for varying probability p of having a missing value. For all three scenarios, we generate data from Clayton copula with parameter 1 and $\text{Exp}(1)$ margins, applying our procedures with $B = 99$, $H = 500$.

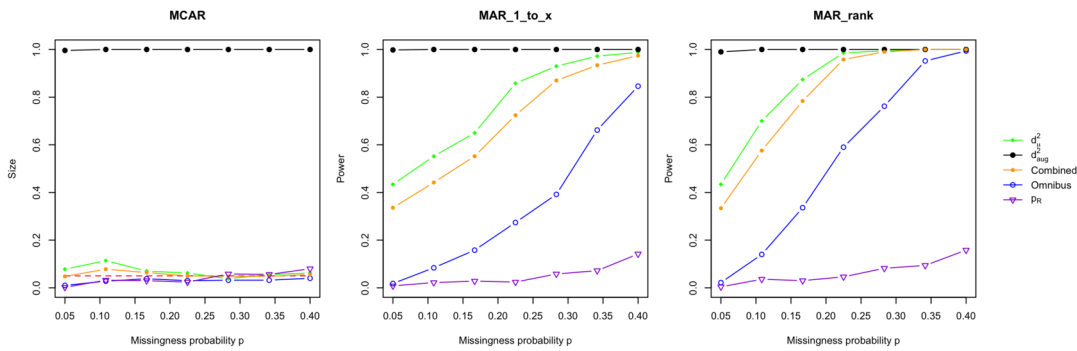


FIG. 7. (Left) Size for MCAR data generated with `delete_MCAR(p)`, for varying probability p of having a missing value; (Centre) Power function for MAR data generated with `delete_MAR_1_to_x(p, x = 9)`, for varying probability p of having a missing value; (Right) Power function for MAR data generated with `delete_MAR_rank(p)`, for varying probability p of having a missing value. For all three scenarios, we generate data from Clayton copula with parameter 1 and $\log N(0, 1)$ margins, applying our procedures with $B = 99$, $H = 500$.

level α , except possibly for Little’s test based on d_{aug}^2 . This outcome supports once more the conjecture in Little (1988) where it is suggested that even under normality, the asymptotic null distribution of d_{aug}^2 is unlikely to be reliable unless the sample size is large. We then move beyond the Gaussian setting and explore different marginal distributions. Specifically, we consider exponential distributions ($\text{Exp}(1)$) in Figure 6, with $d = 3$, and log-normal distributions ($\log N(0, 1)$) in Figure 7, with $d = 5$. In the setting of log-normal data, the centre and right panels show a slight loss in power for our omnibus and combined approaches. However, this is offset by their better control of the Type-I error (left panel) compared to Little’s test based on d_μ^2 , and especially d_{aug}^2 , which consistently fails to recognize the null and always rejects. The results for the exponential setting fall between those of the Gaussian and log-normal settings.

Overall, these simulations demonstrate the effectiveness of our procedures. The combined approach could serve as a valid extension of Little’s test based on d_μ^2 when both means and covariance matrices are taken into account, rather than just means. Interestingly, incorporating d_μ^2 with Algorithms 1 and 3 does not inflate the Type-I error, which is a significant limitation of Little’s test based on d_{aug}^2 . On the other hand, while the omnibus approach exhibits slightly weaker performance in terms of power, it provides more rigorous control of the Type-I error. Additionally, it offers two further advantages compared to the combined approach: (i) it can be applied to any missingness pattern $\mathbb{S}$, even when not all pairs of variables are observed simultaneously; and (ii) it remains effective even when the dimensionality d is large, as it does not depend on the EM algorithm, which is known to encounter issues in practice when $d > 50$ (see the documentation of the R function `na.test` in Yanagida (2024)).

Acknowledgments. We thank the anonymous reviewers for their constructive comments, which helped improving the quality of the paper.

Funding. The second author was supported by Engineering and Physical Sciences Research Council (EPSRC) New Investigator Award EP/W016117/1 and European Research Council Starting Grant 101163546.

SUPPLEMENTARY MATERIAL

Supplementary Material A: Tests of missing completely at random based on sample covariance matrices (DOI: [10.1214/25-AOS2540SUPPA](https://doi.org/10.1214/25-AOS2540SUPPA); .pdf). The supplement Bordino and Berrett (2025) contains proofs of our main results, as well as some auxiliary results.

Supplementary Material B: Tests of missing completely at random based on sample covariance matrices (DOI: [10.1214/25-AOS2540SUPPB](https://doi.org/10.1214/25-AOS2540SUPPB); .zip). This supplement contains the code to reproduce the simulation studies presented in Section 6.

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